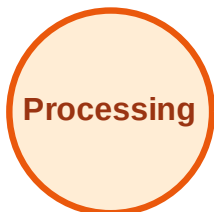


# BFS Traversal

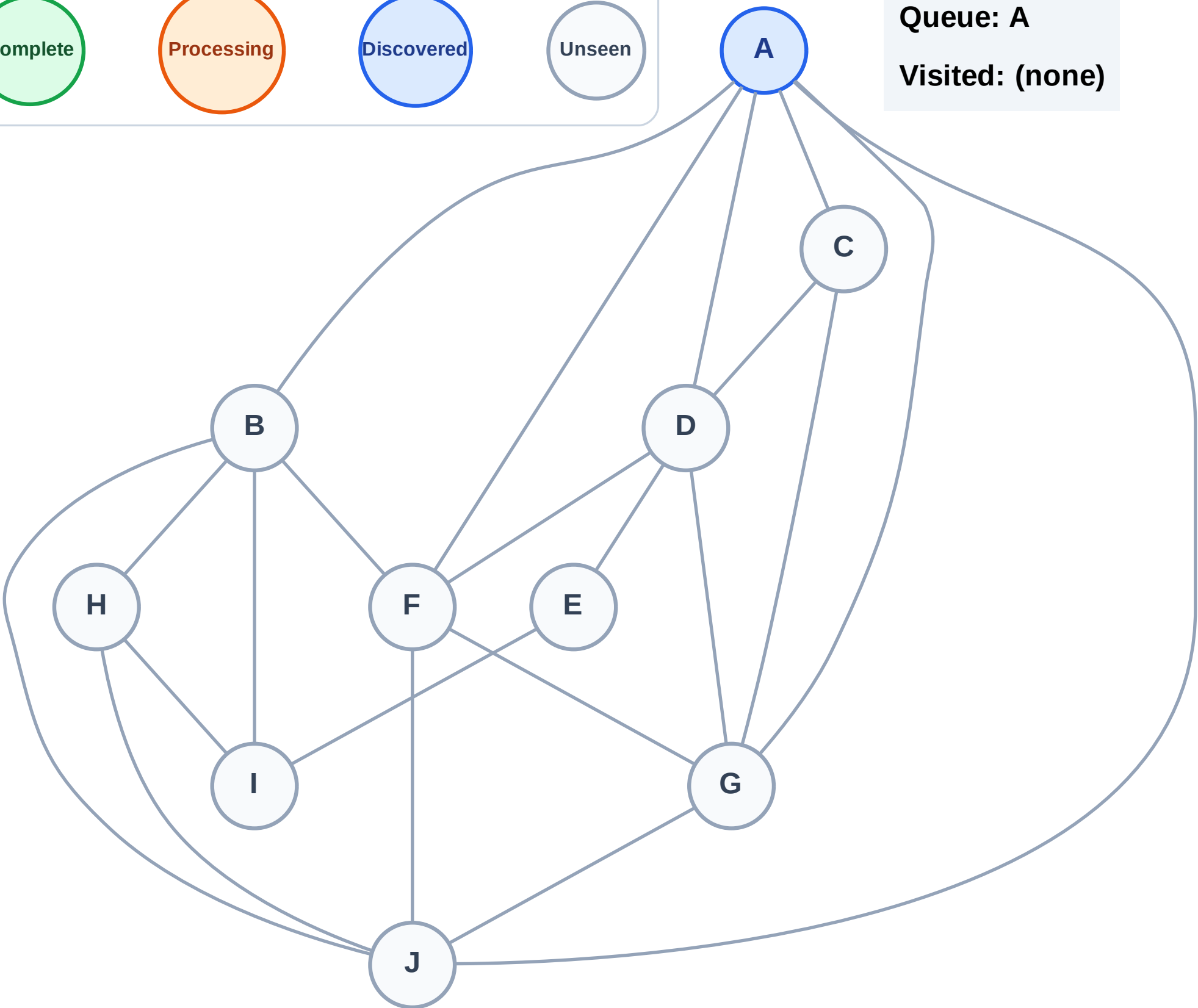
Step 1: Start at A

## Legend



Queue: A

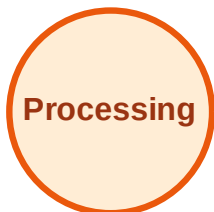
Visited: (none)



# BFS Traversal

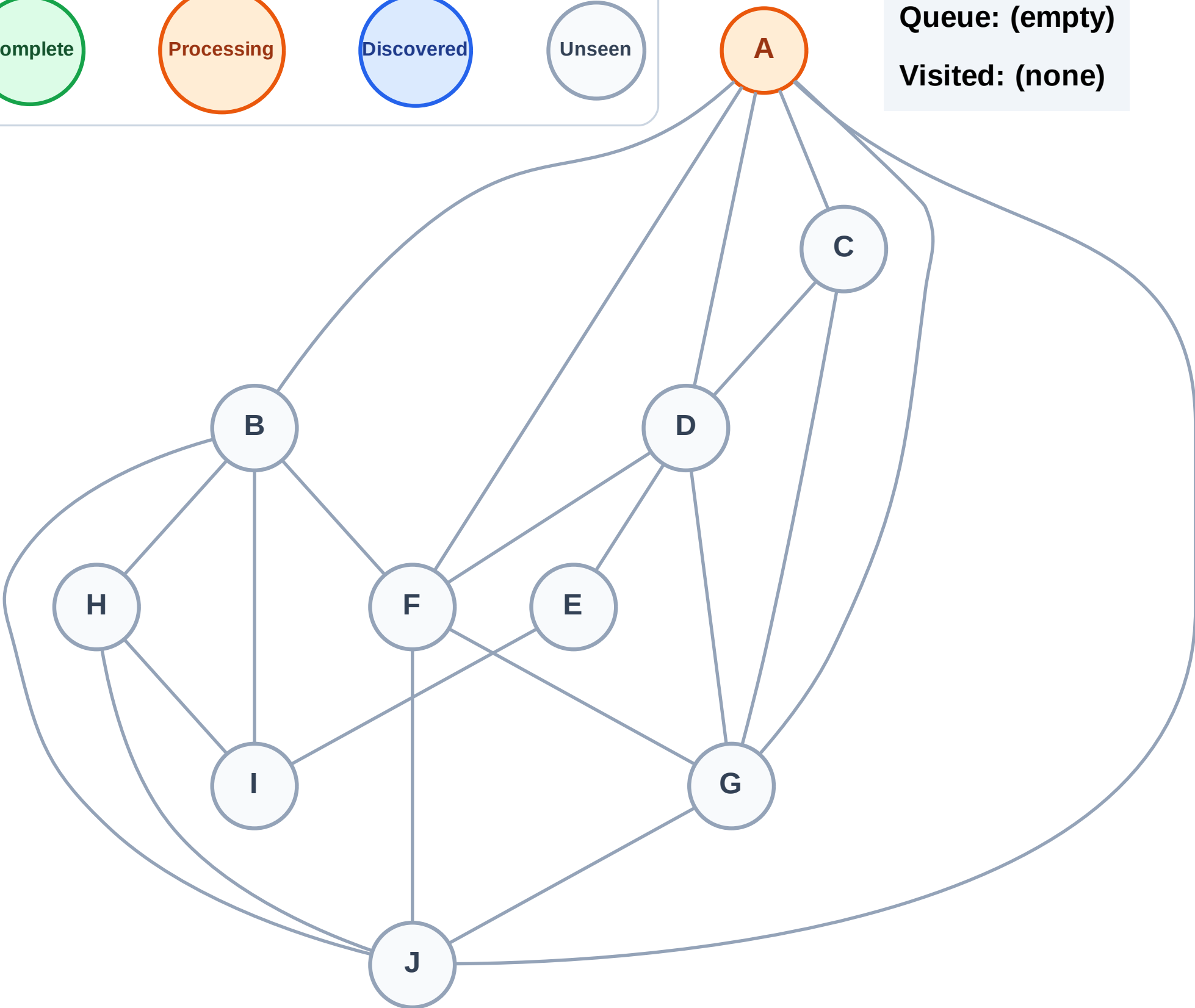
Step 2: Process node A

## Legend



Queue: (empty)

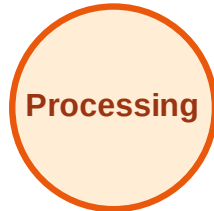
Visited: (none)



# BFS Traversal

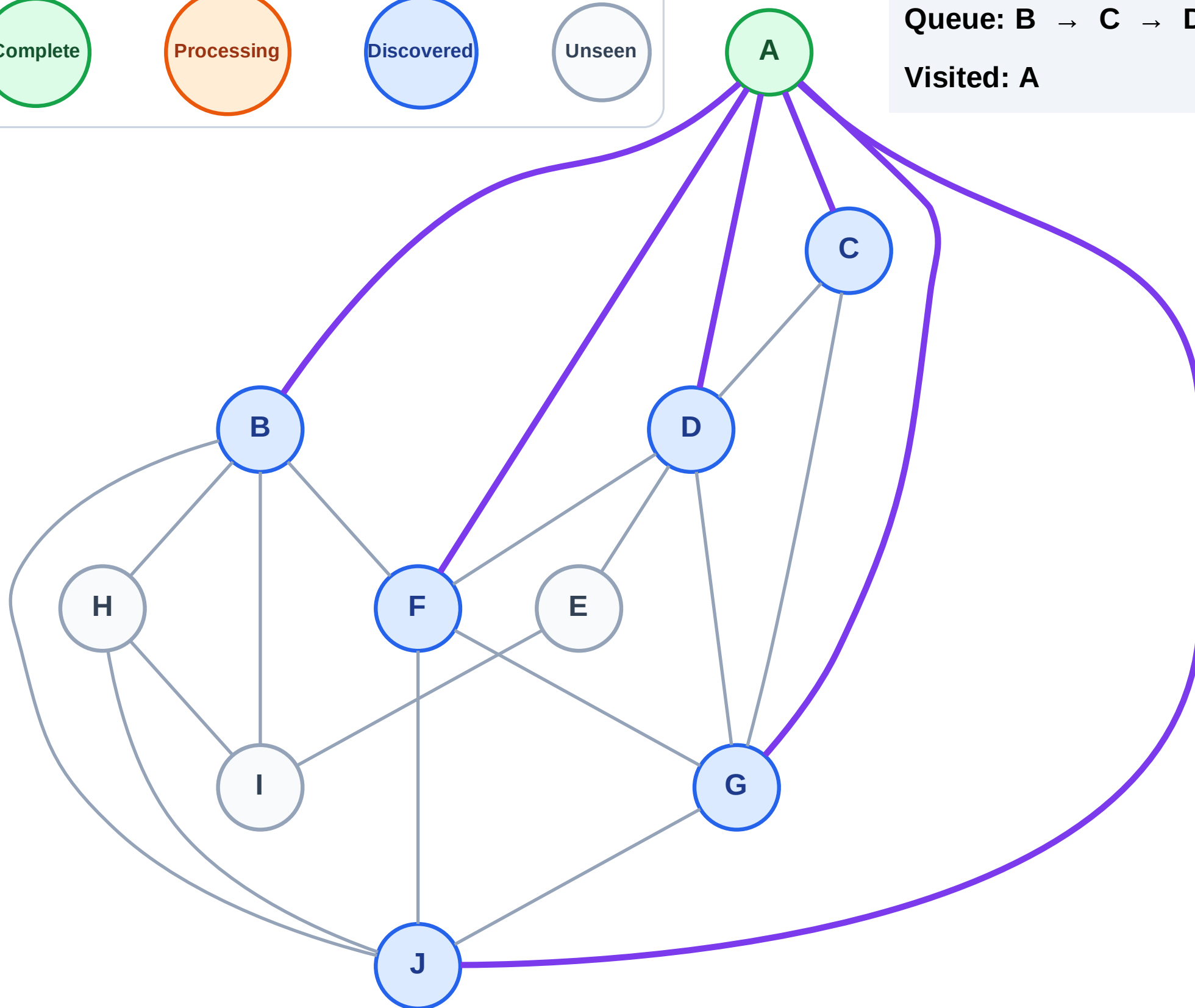
Step 3: Discover B, C, D, F, G, J from A

## Legend



Queue: B → C → D → F → G → J

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

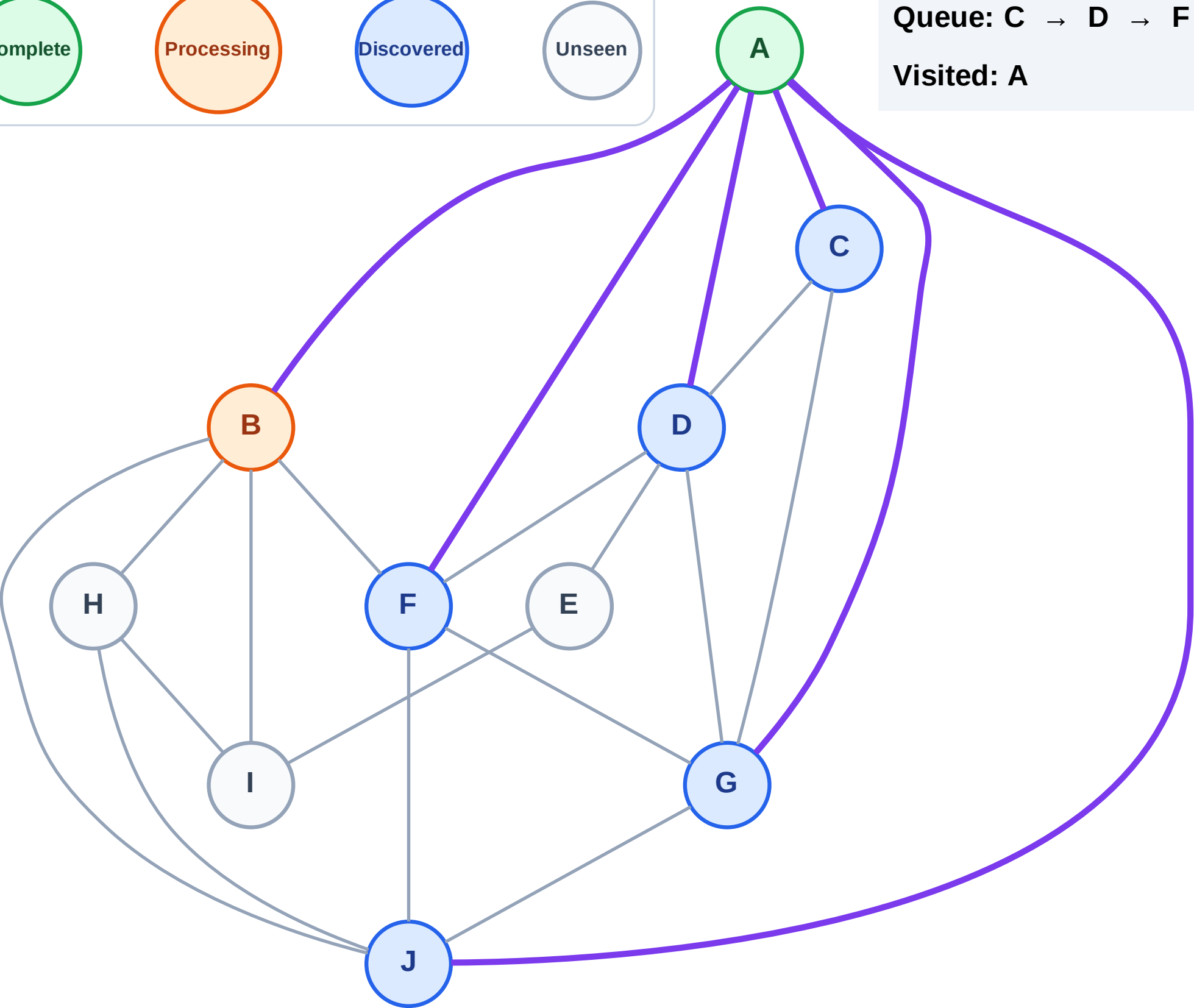
Processing

Discovered

Unseen

Queue: C → D → F → G → J

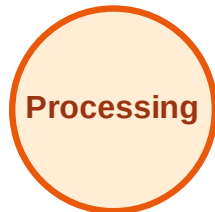
Visited: A



# BFS Traversal

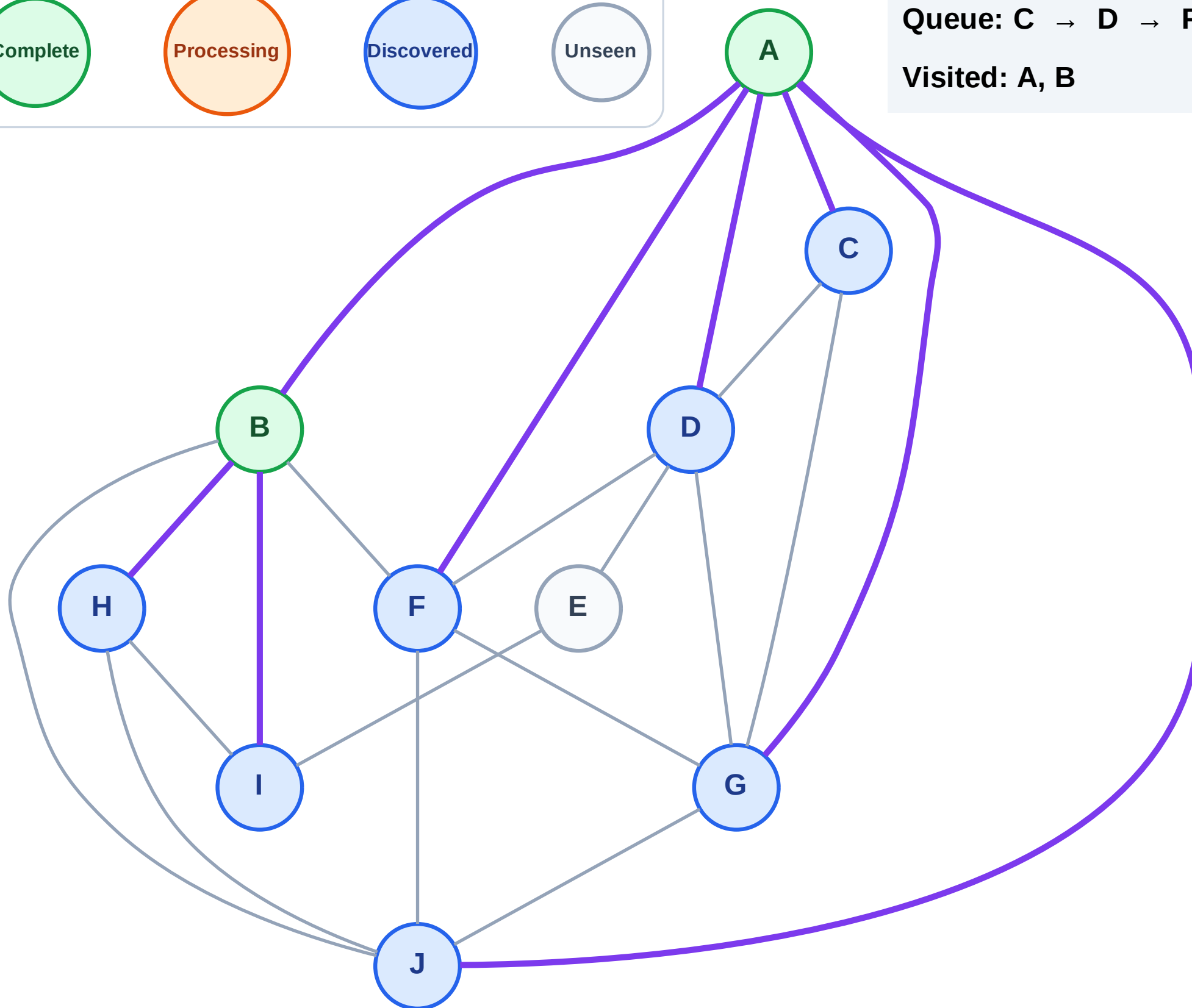
Step 5: Discover H, I from B

## Legend



Queue: C → D → F → G → J → H → I

Visited: A, B



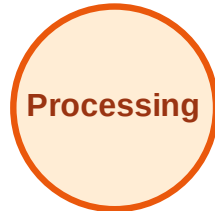
# BFS Traversal

Step 6: Process node C

## Legend



Complete



Processing



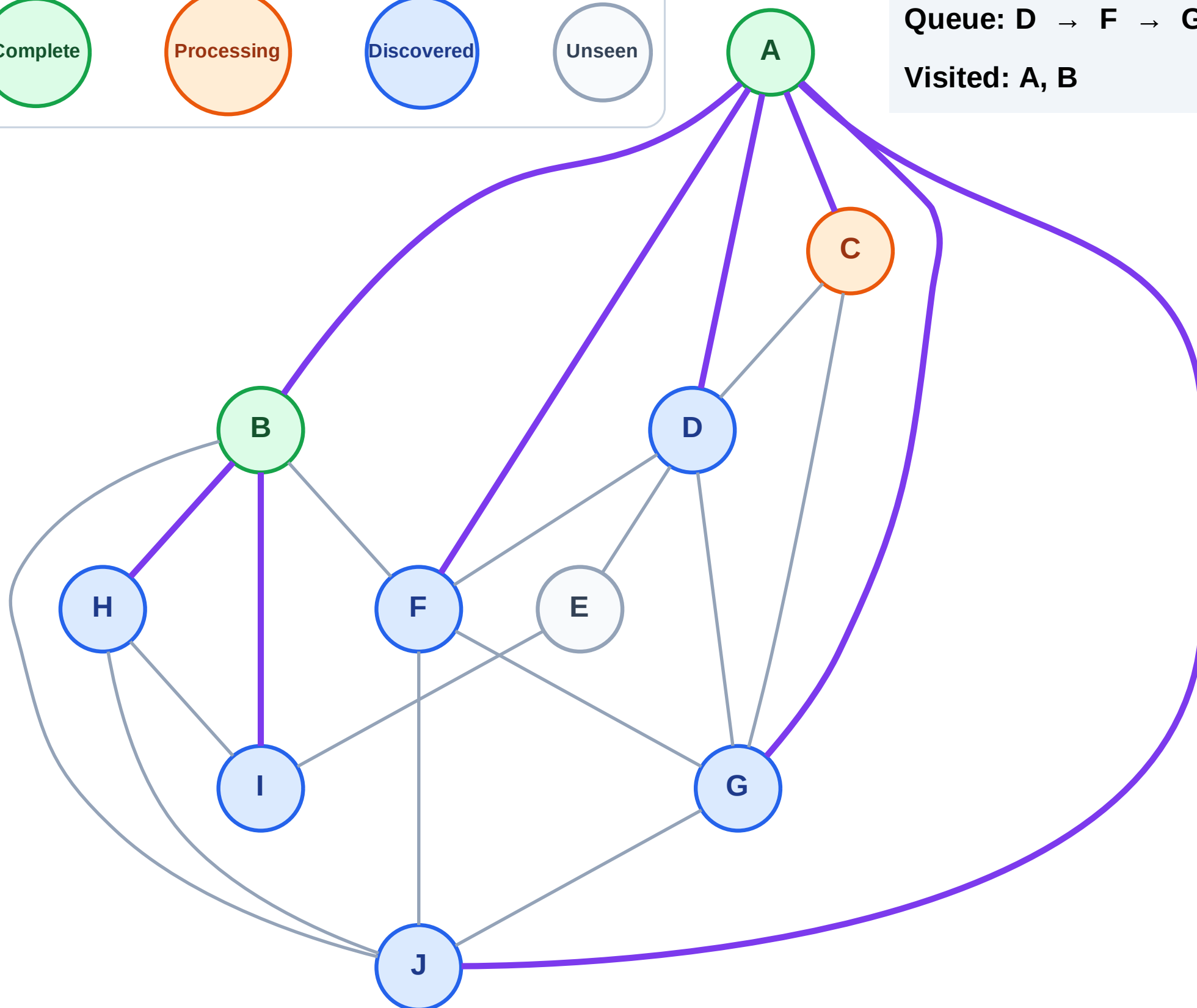
Discovered



Unseen

Queue: D → F → G → J → H → I

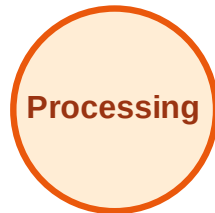
Visited: A, B



# BFS Traversal

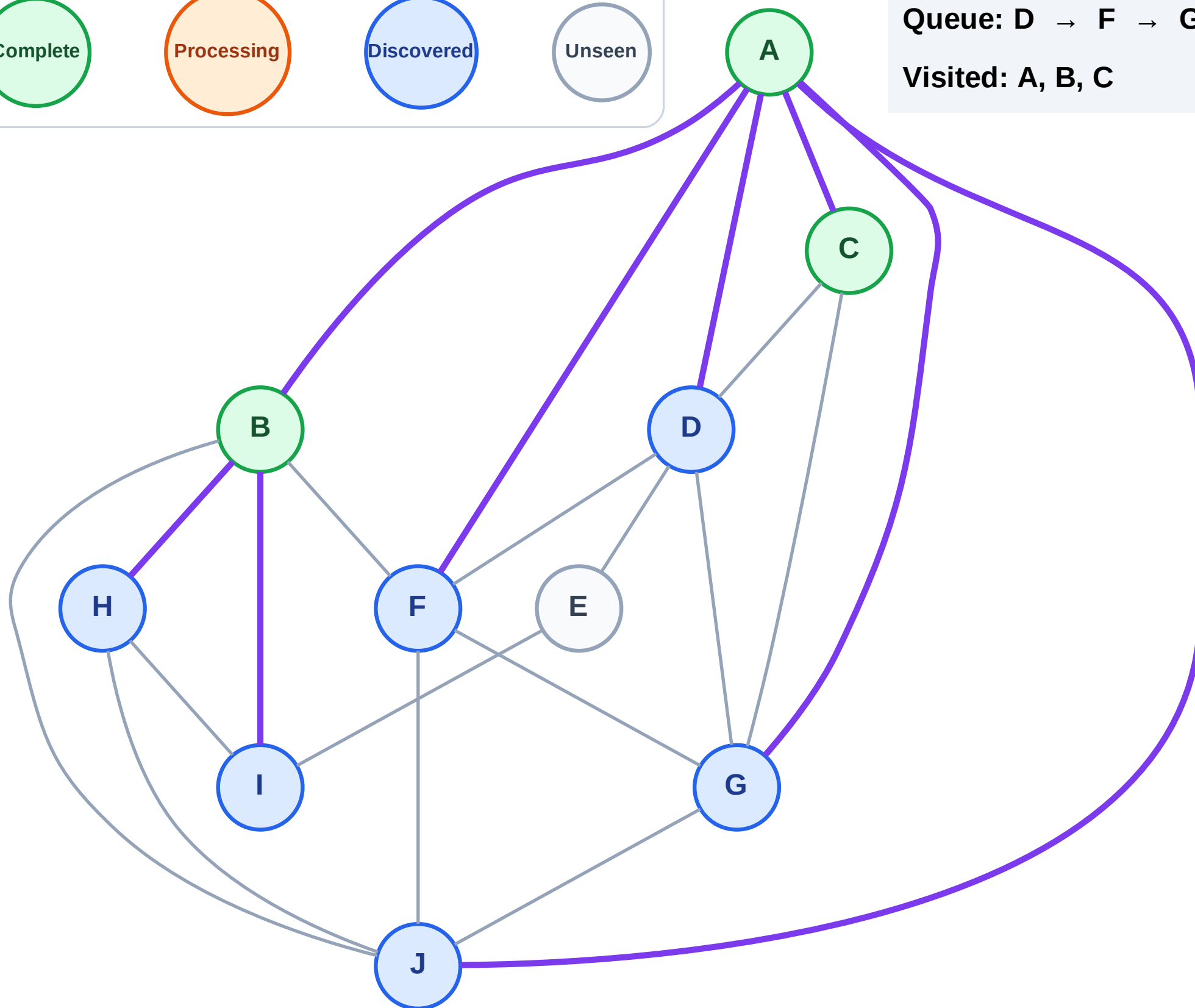
Step 7: No new neighbors from C

## Legend



Queue: D → F → G → J → H → I

Visited: A, B, C





# BFS Traversal

Step 8: Process node D

Legend

Complete

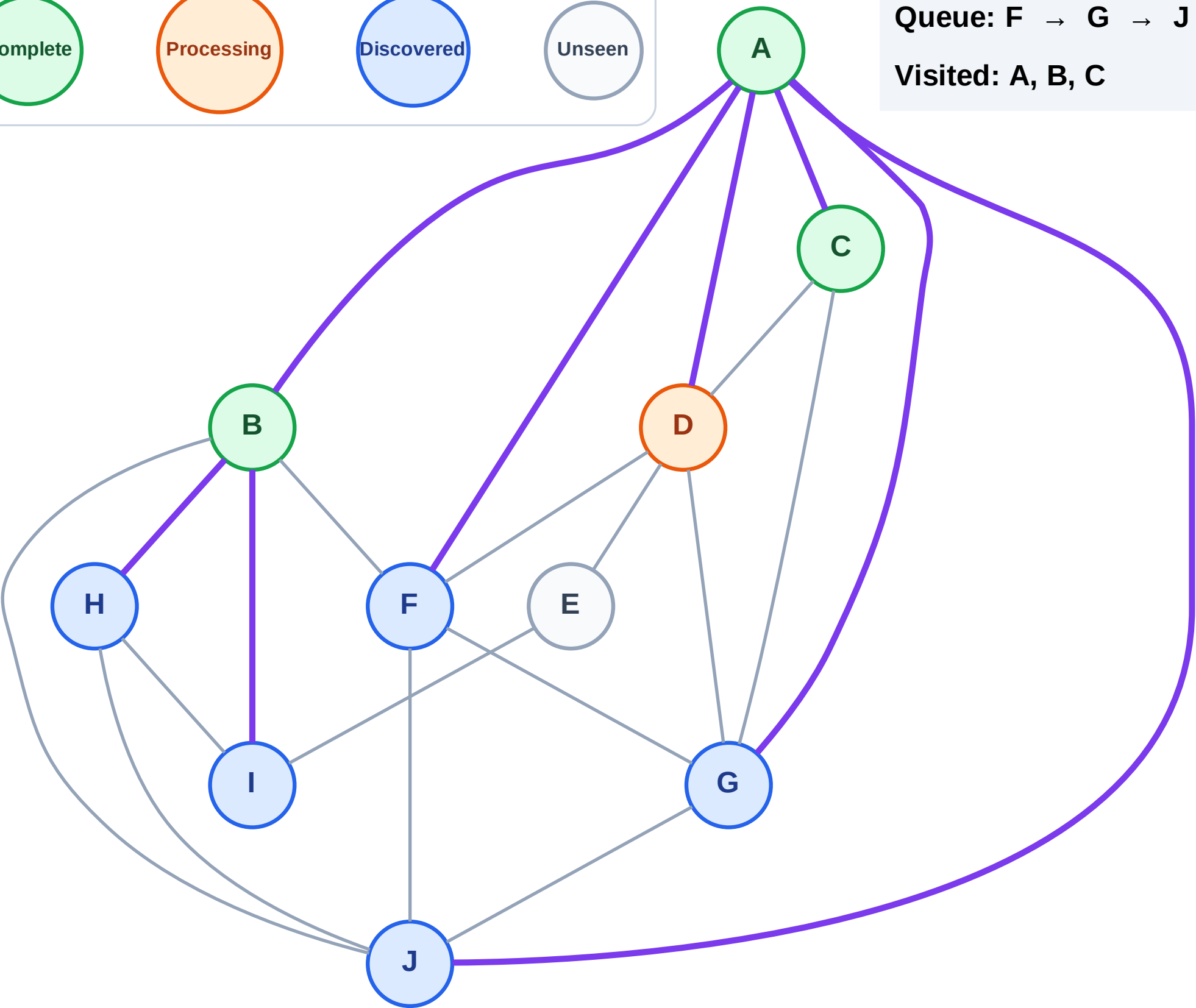
Processing

Discovered

Unseen

Queue: F → G → J → H → I

Visited: A, B, C

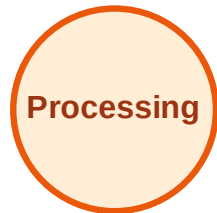




# BFS Traversal

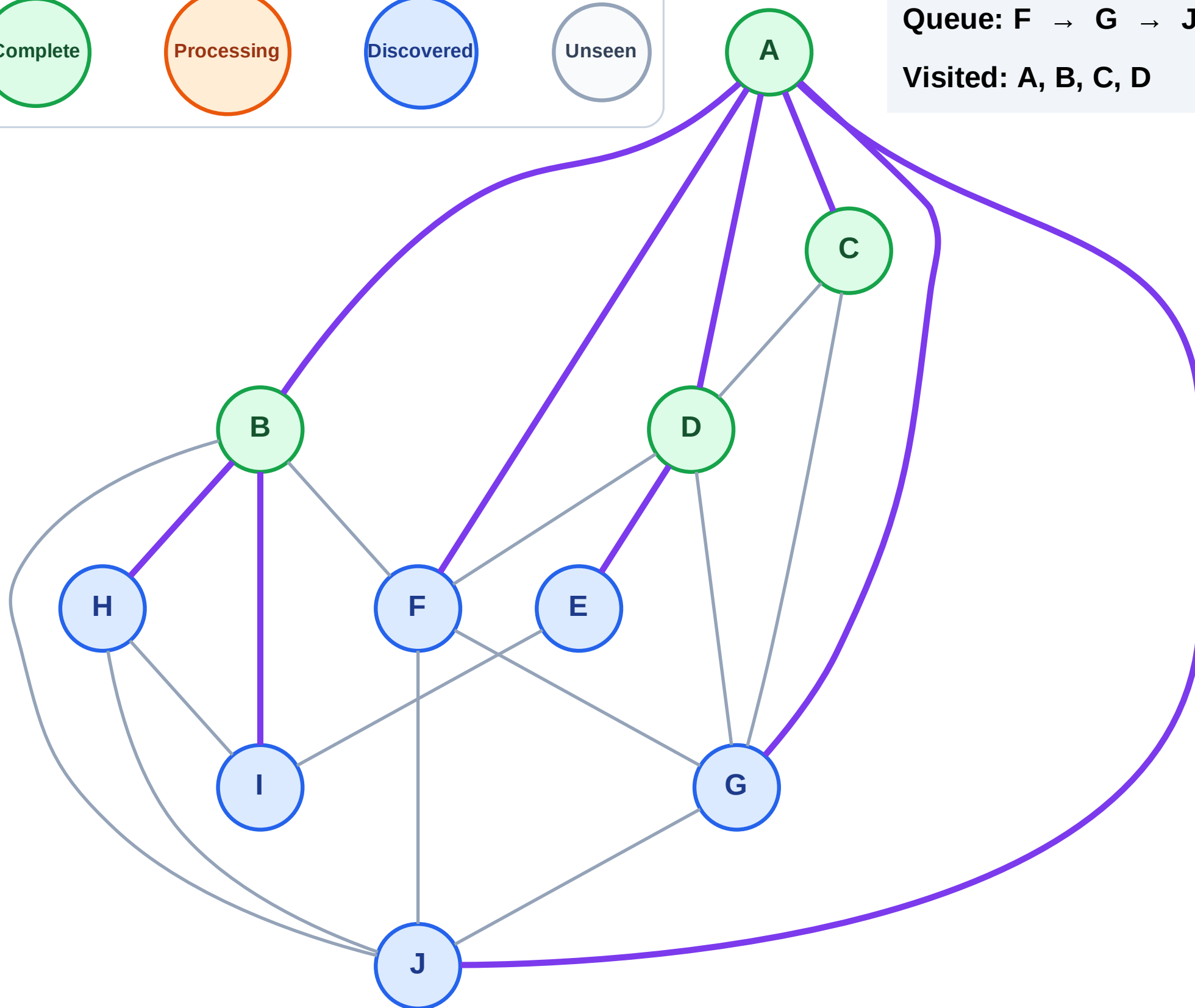
Step 9: Discover E from D

## Legend



Queue: F → G → J → H → I → E

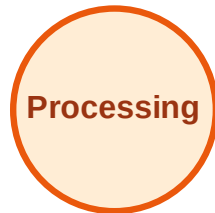
Visited: A, B, C, D



# BFS Traversal

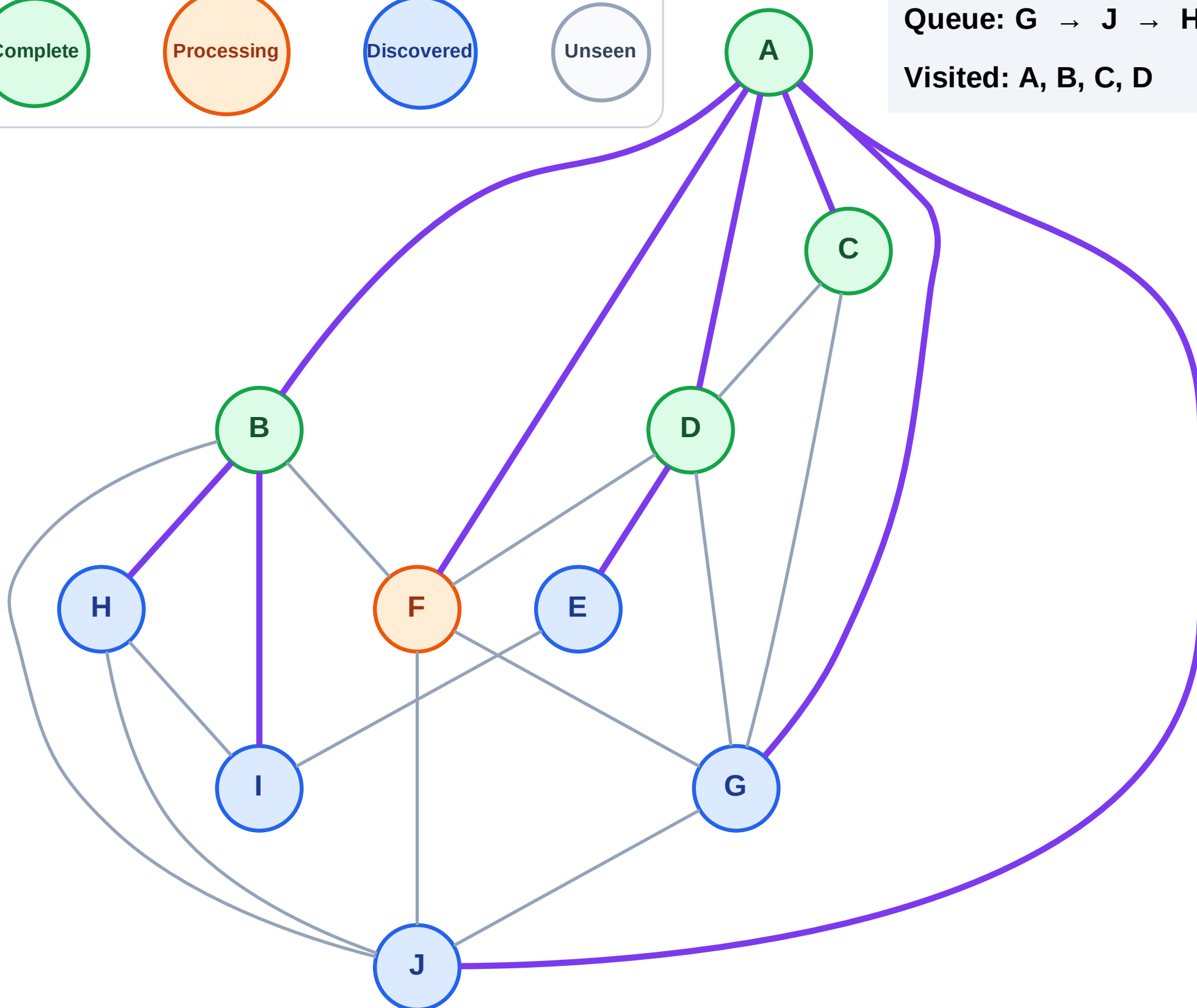
Step 10: Process node F

## Legend



Queue: G → J → H → I → E

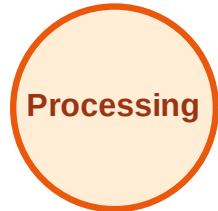
Visited: A, B, C, D



# BFS Traversal

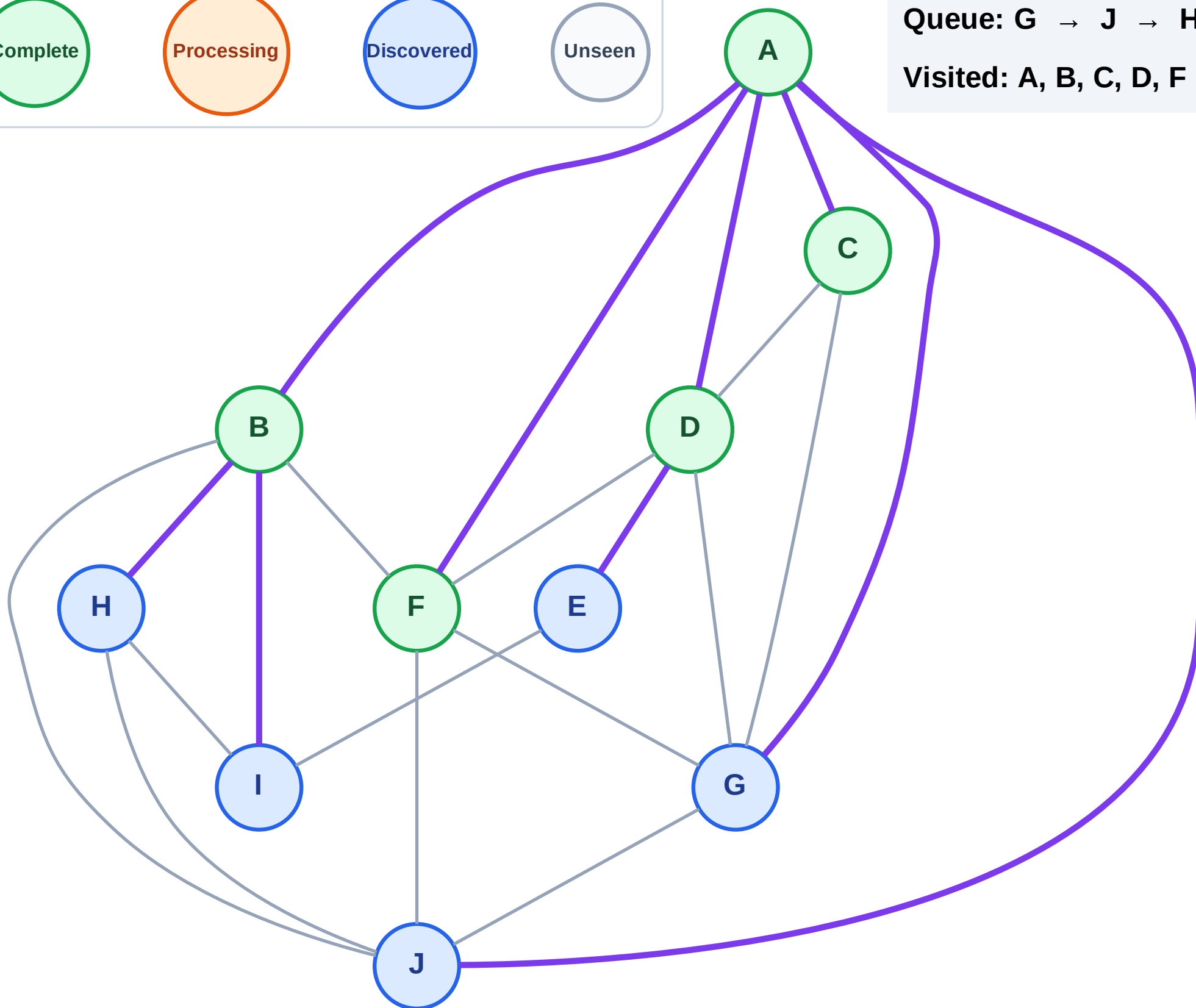
Step 11: No new neighbors from F

## Legend



Queue: G → J → H → I → E

Visited: A, B, C, D, F



# BFS Traversal

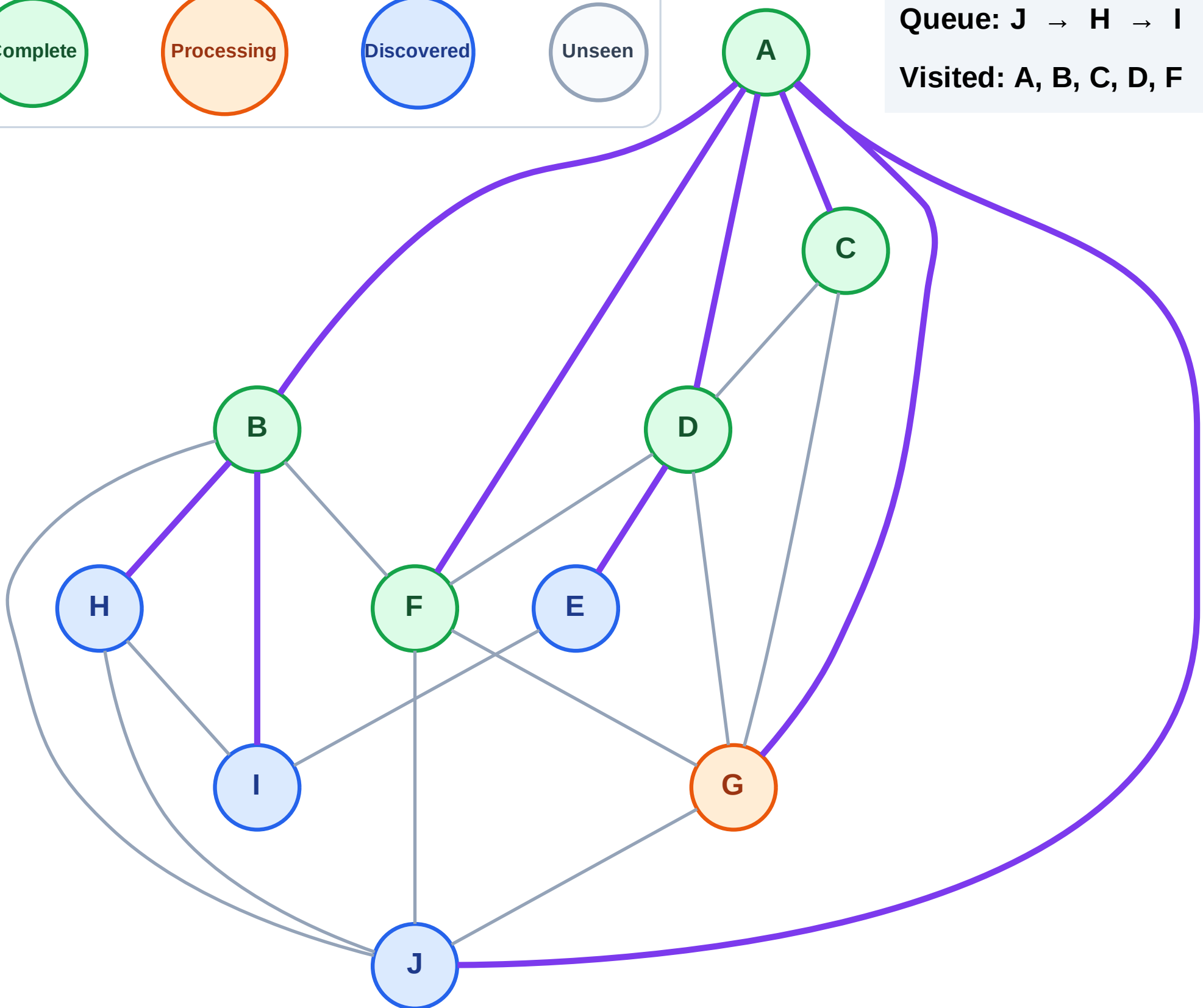
Step 12: Process node G

## Legend



Queue: J → H → I → E

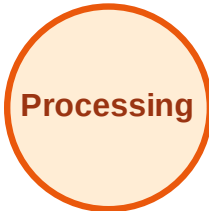
Visited: A, B, C, D, F



# BFS Traversal

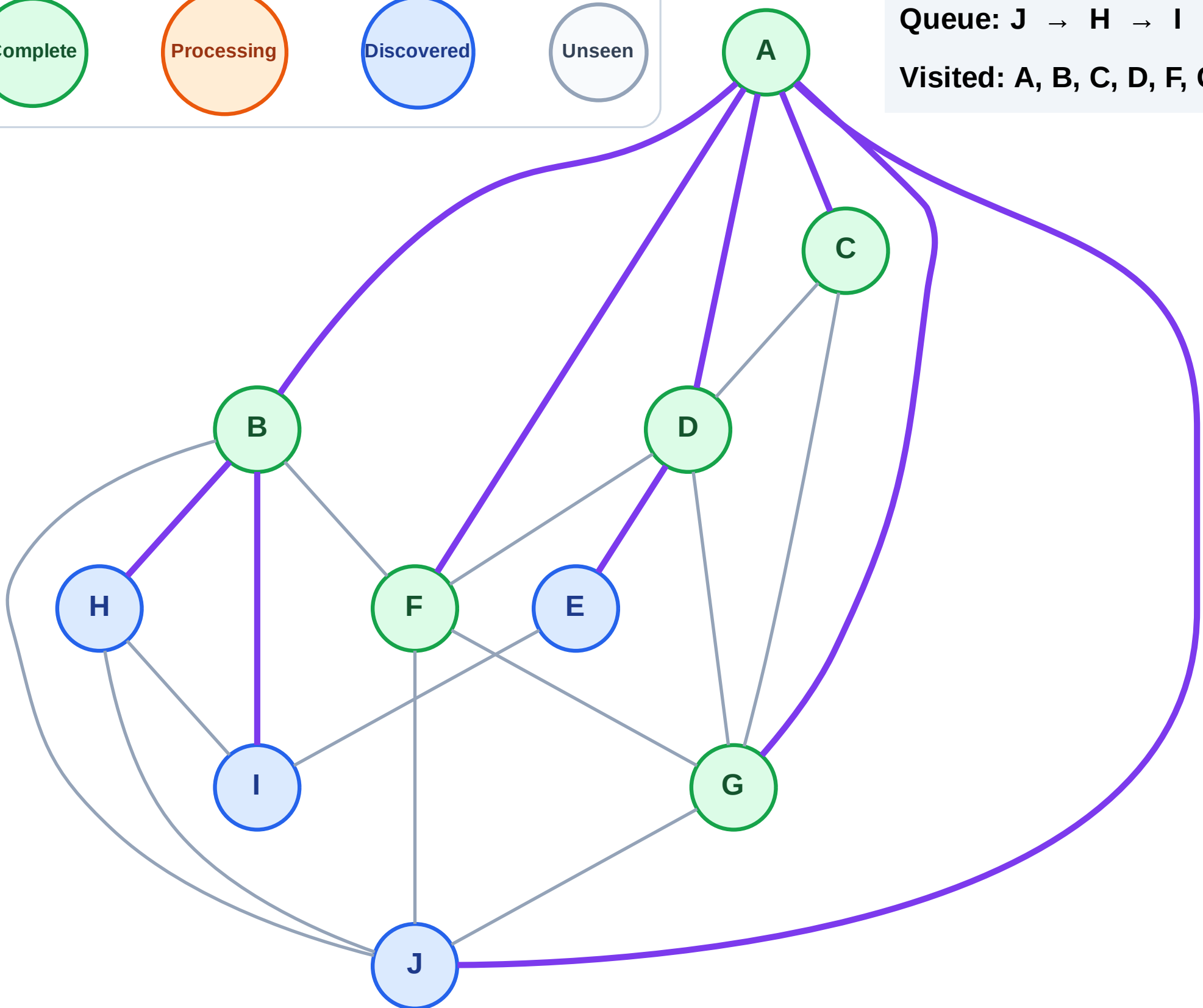
Step 13: No new neighbors from G

## Legend



Queue: J → H → I → E

Visited: A, B, C, D, F, G



# BFS Traversal

Step 14: Process node J

Legend

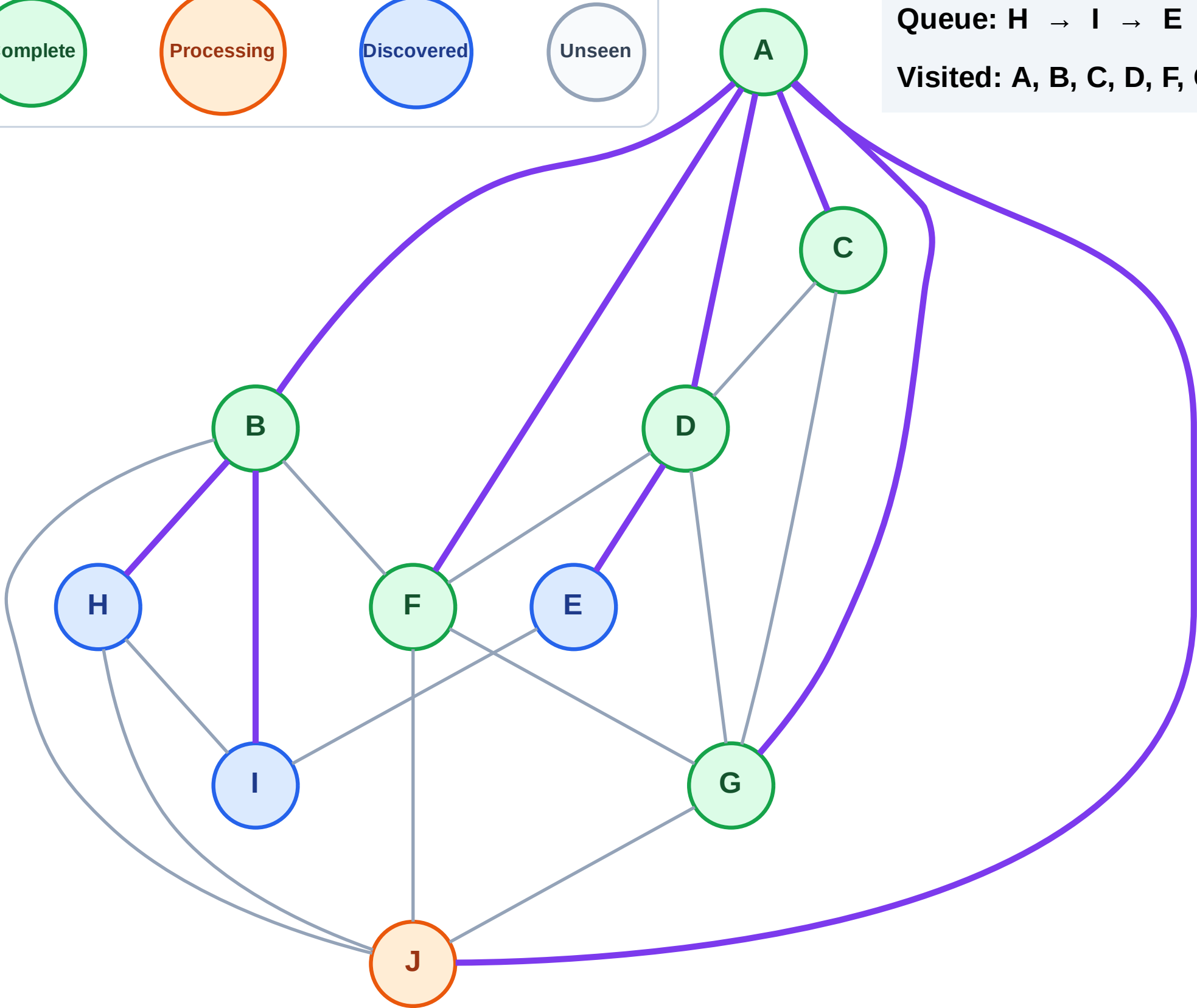
Complete

Processing

Discovered

Unseen

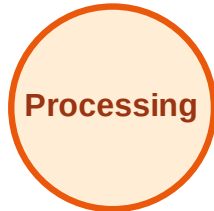
Queue: H → I → E  
Visited: A, B, C, D, F, G



# BFS Traversal

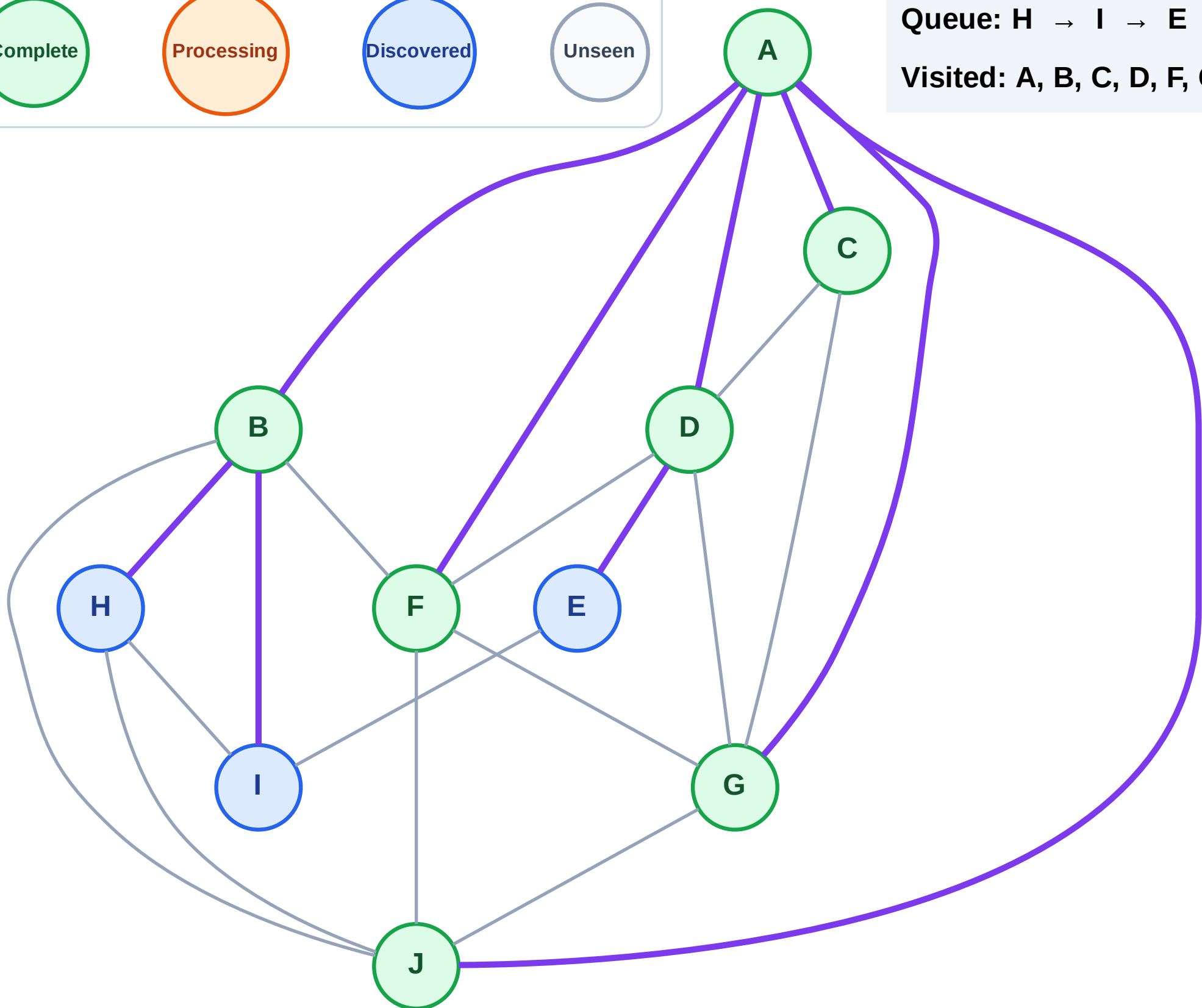
Step 15: No new neighbors from J

## Legend



Queue: H → I → E

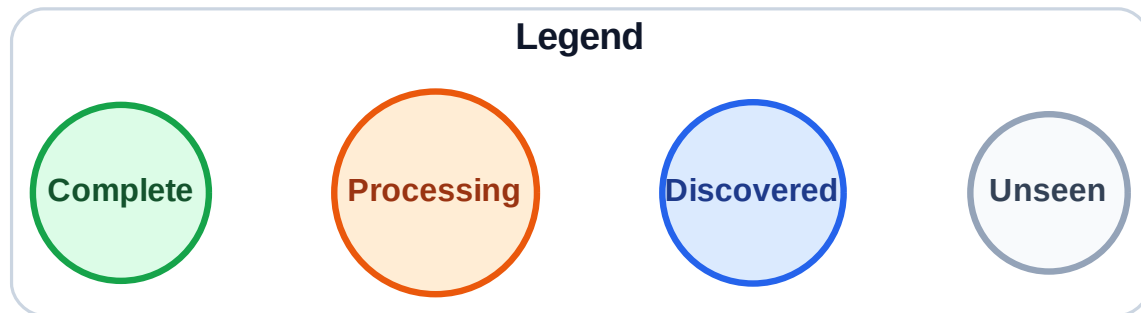
Visited: A, B, C, D, F, G, J





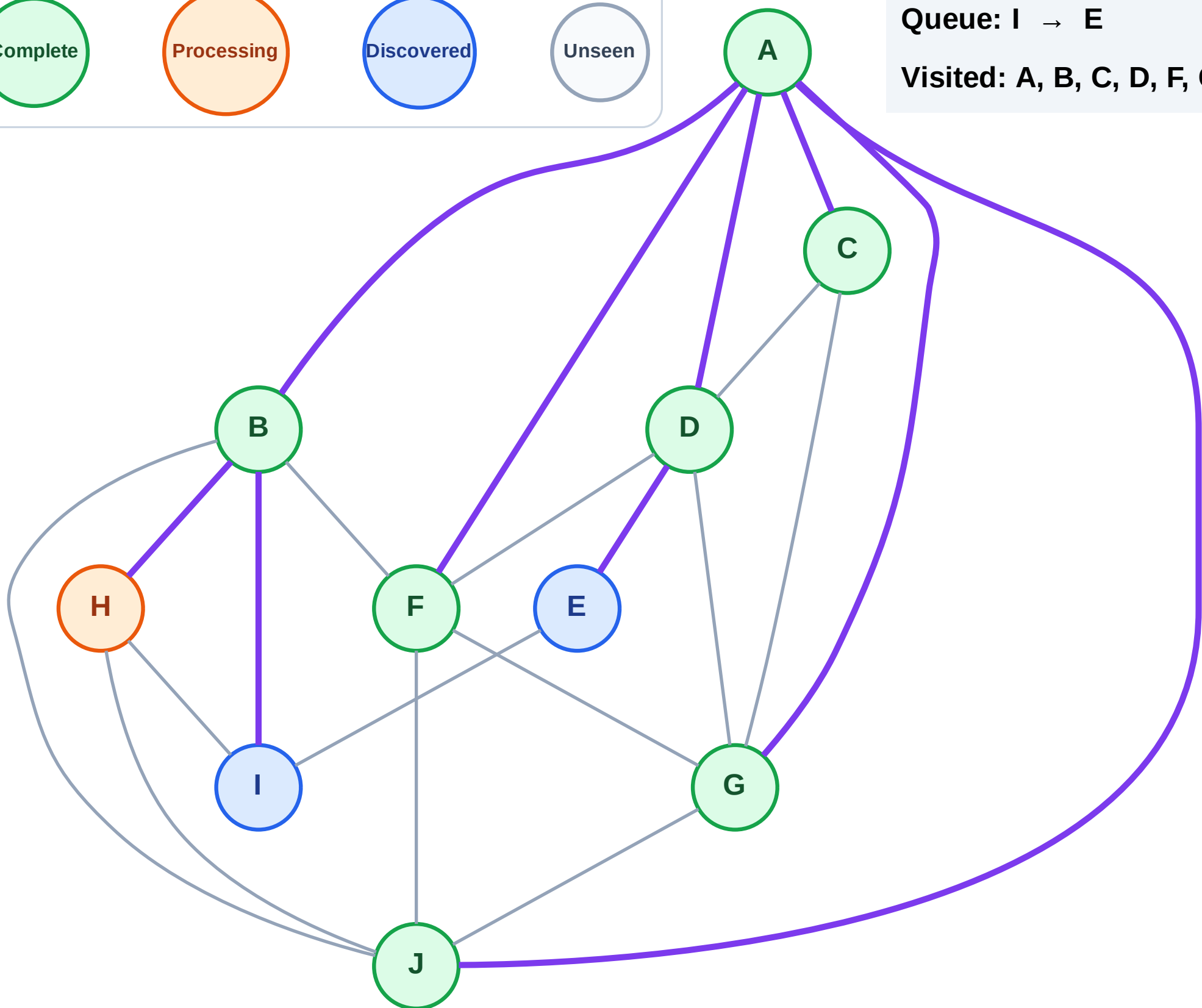
### Step 16: Process node H

### Step 16: Process node H



**Queue: I → E**

**Visited: A, B, C, D, F, G, J**

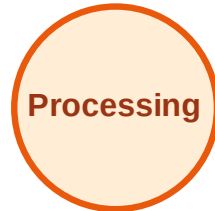




# BFS Traversal

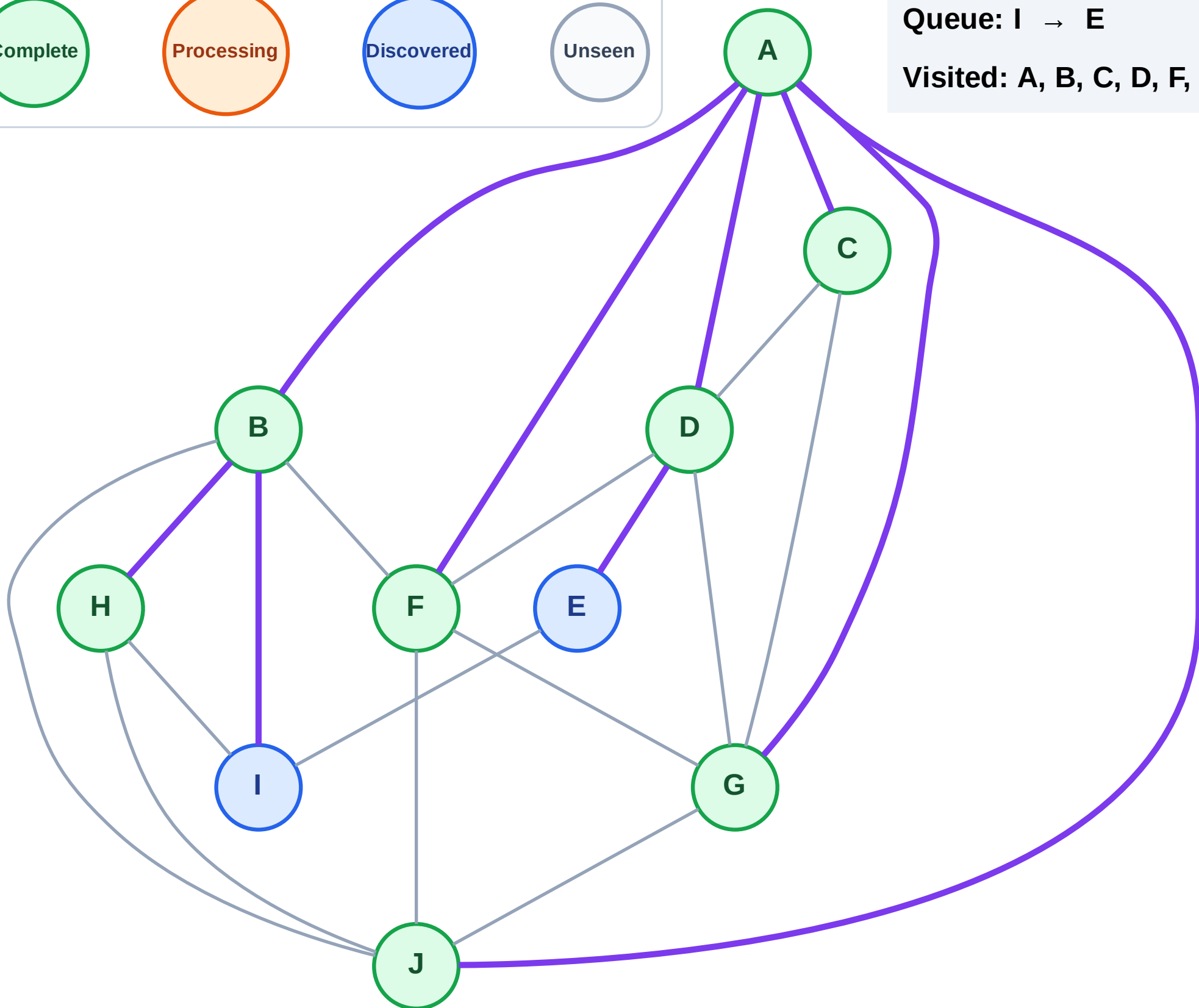
Step 17: No new neighbors from H

## Legend



Queue: I → E

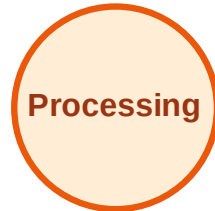
Visited: A, B, C, D, F, G, H, J



# BFS Traversal

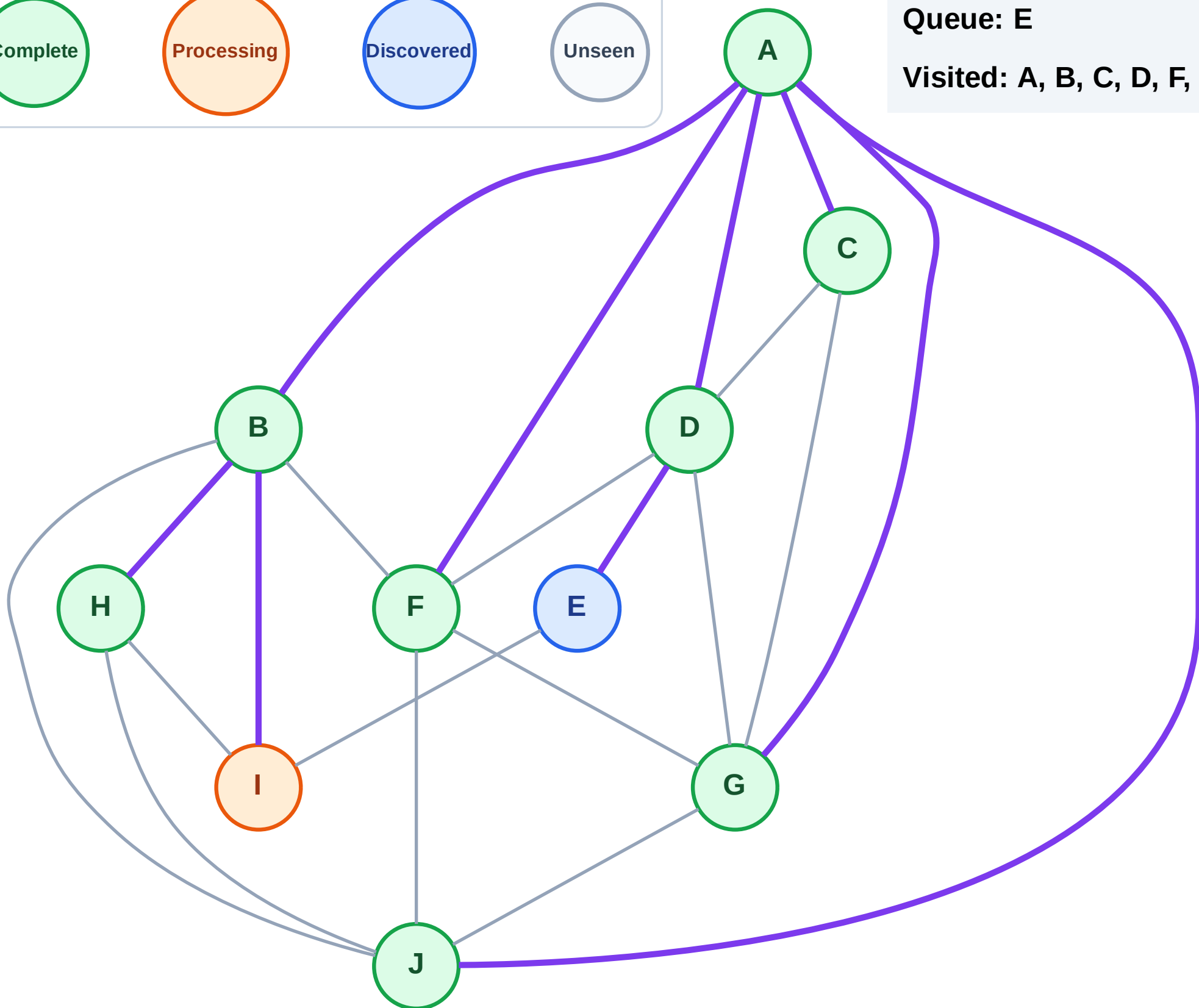
Step 18: Process node I

## Legend



Queue: E

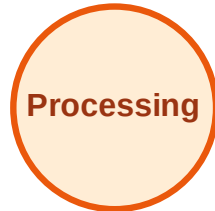
Visited: A, B, C, D, F, G, H, J



# BFS Traversal

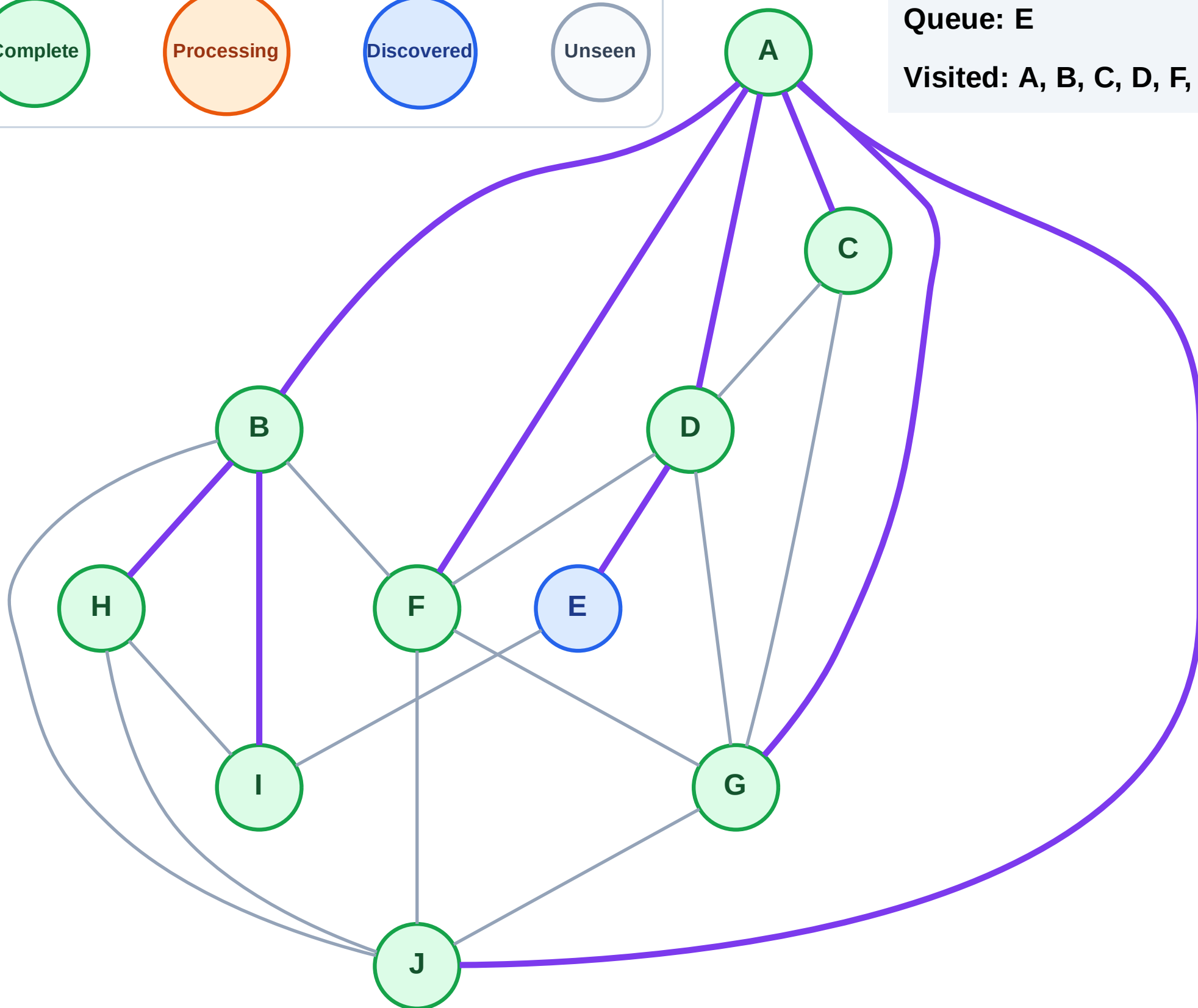
Step 19: No new neighbors from I

## Legend



Queue: E

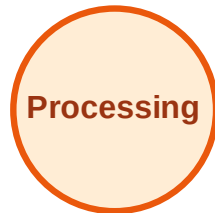
Visited: A, B, C, D, F, G, H, I, J



# BFS Traversal

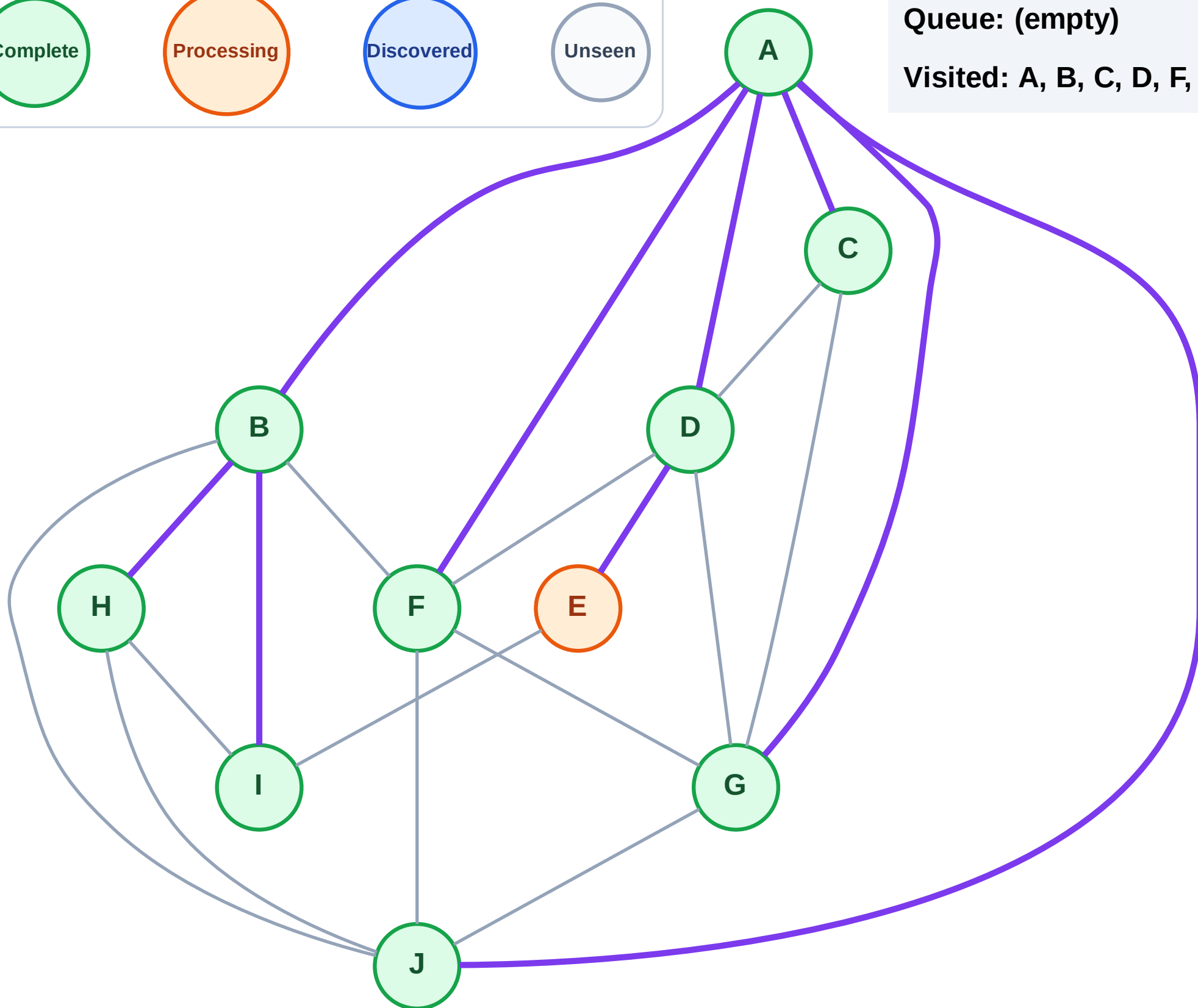
Step 20: Process node E

## Legend



Queue: (empty)

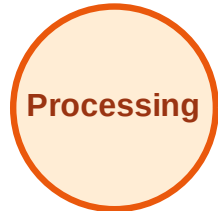
Visited: A, B, C, D, F, G, H, I, J



# BFS Traversal

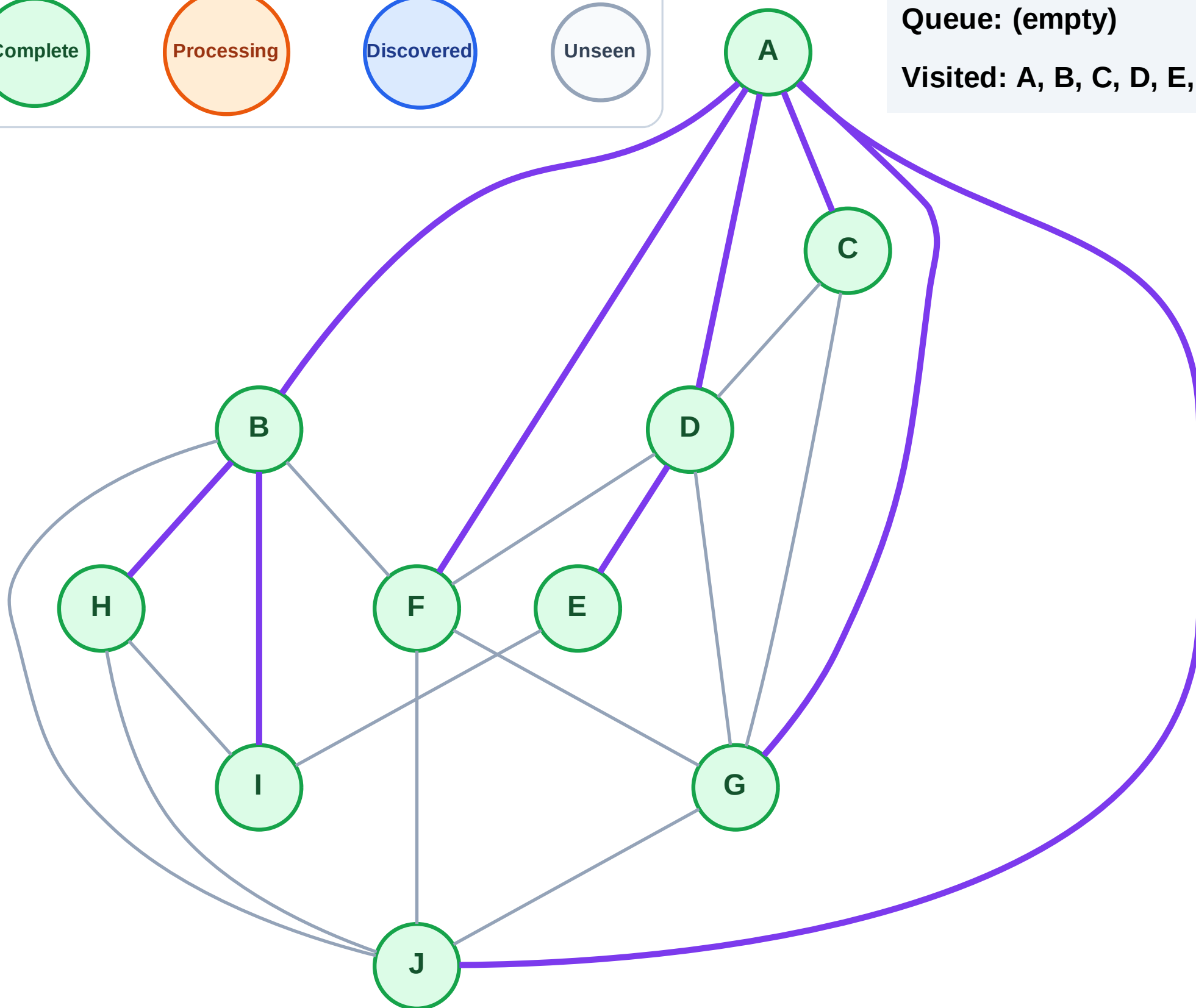
Step 21: No new neighbors from E

## Legend



Queue: (empty)

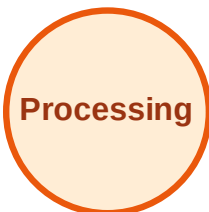
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

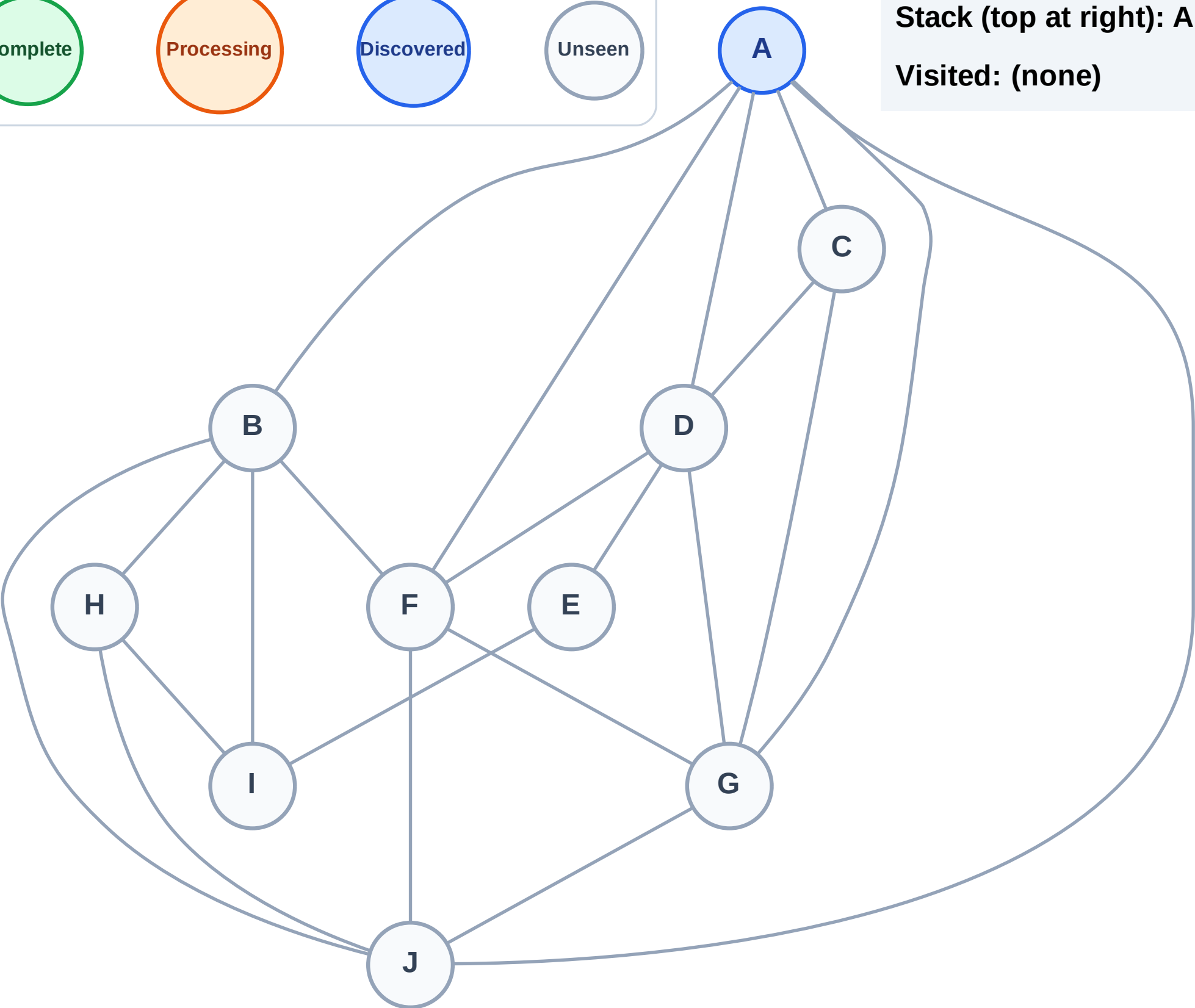
Step 1: Start at A

## Legend



Stack (top at right): A

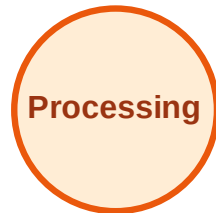
Visited: (none)



# DFS Traversal

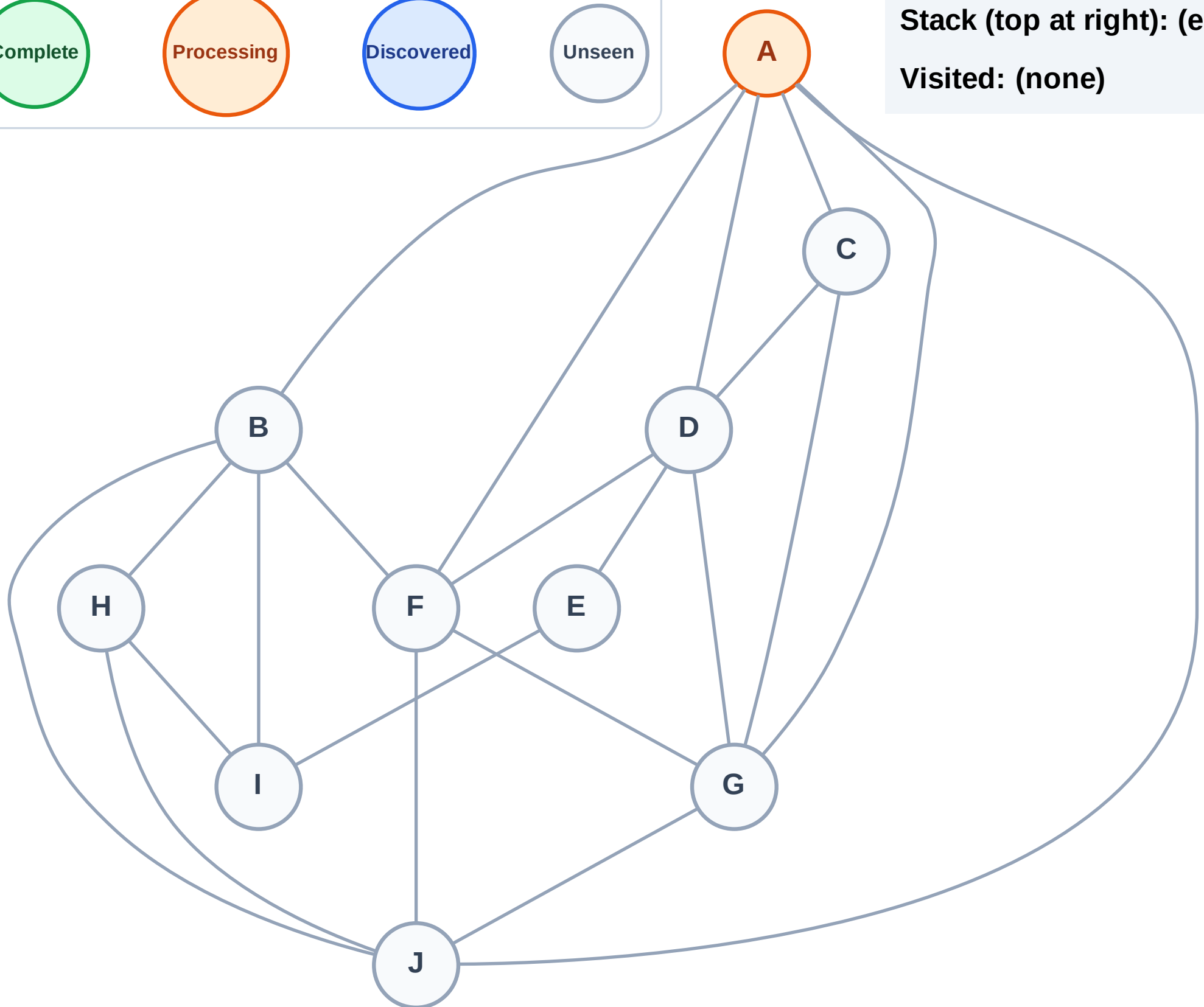
Step 2: Process node A

## Legend



Stack (top at right): (empty)

Visited: (none)

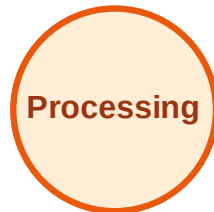




# DFS Traversal

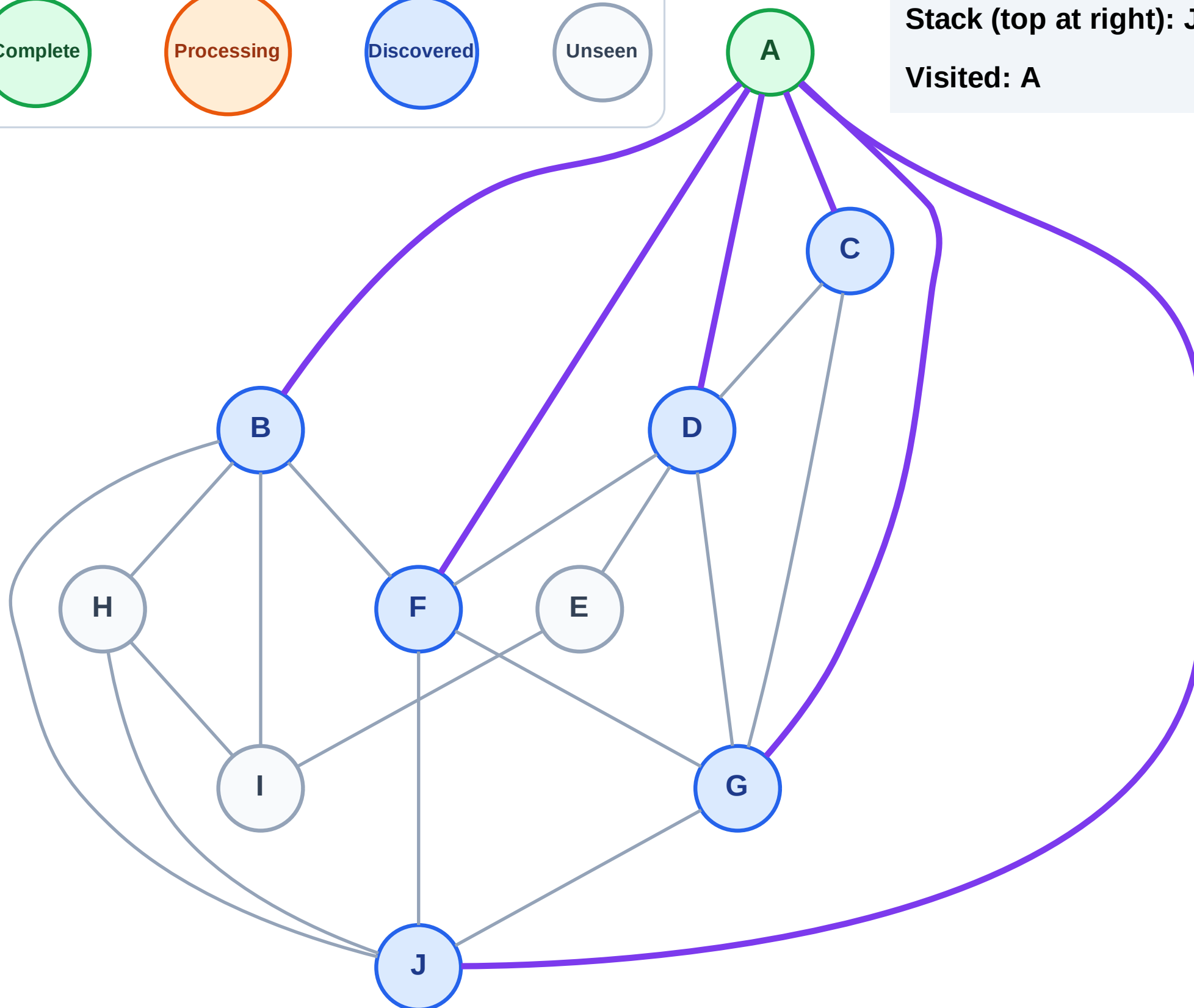
Step 3: Discover J, G, F, D, C, B from A

## Legend



Stack (top at right): J | G | F | D | C | B

Visited: A





# DFS Traversal

Step 4: Process node B

## Legend



Complete



Processing



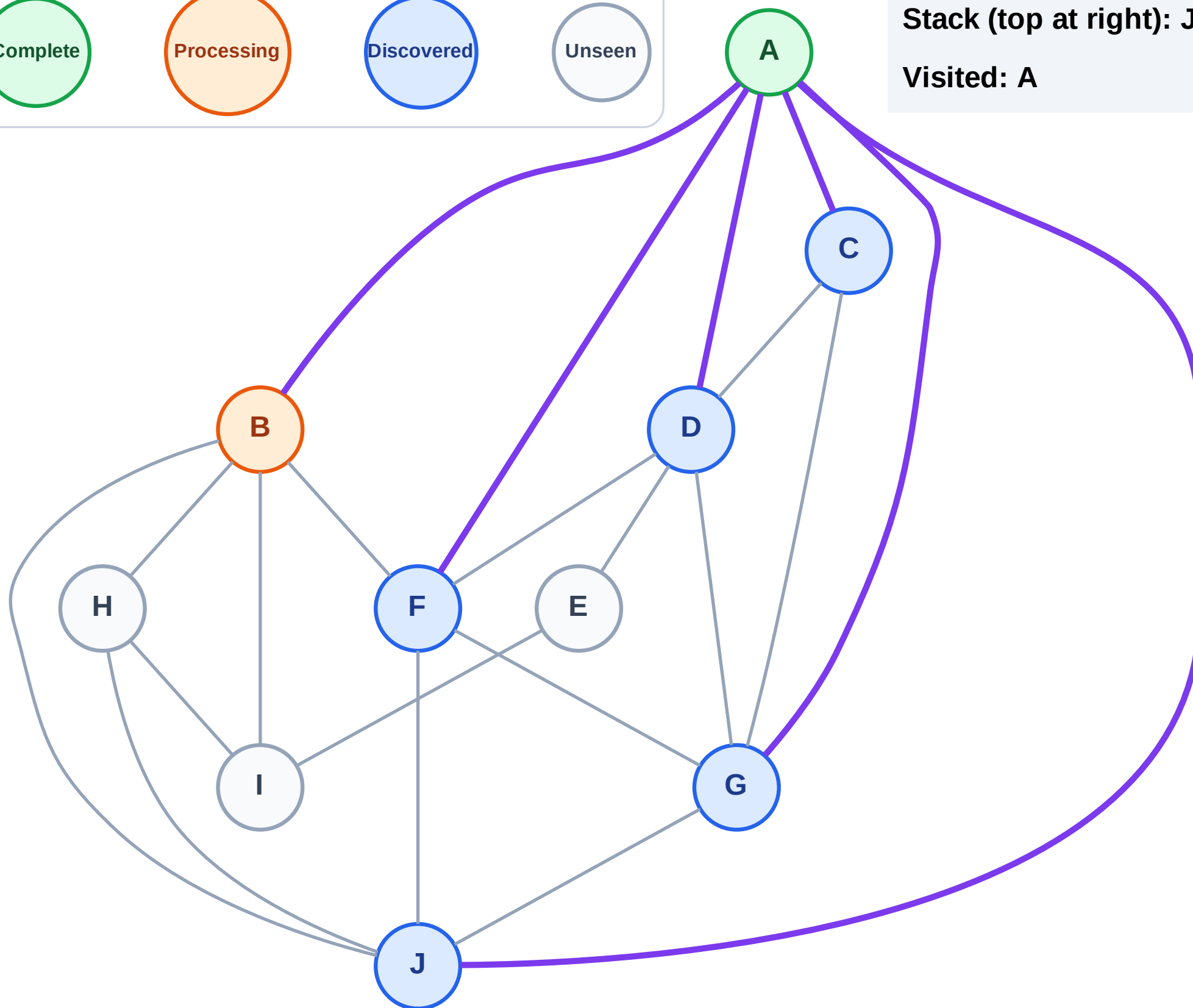
Discovered



Unseen

Stack (top at right): J | G | F | D | C

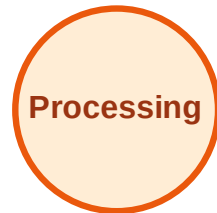
Visited: A



# DFS Traversal

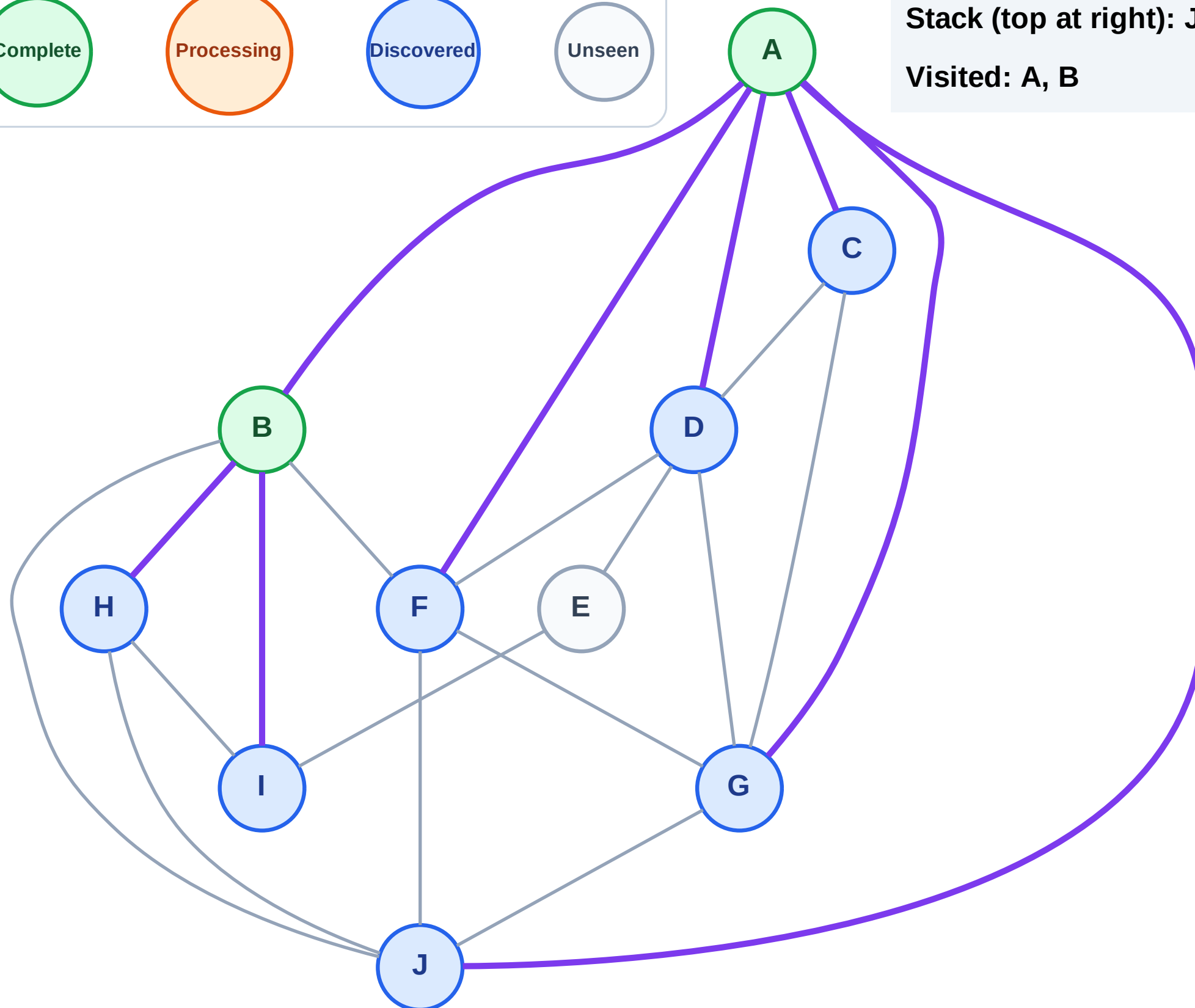
Step 5: Discover I, H from B

## Legend



Stack (top at right): J | G | F | D | C | I | H

Visited: A, B



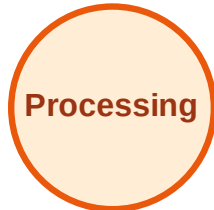
# DFS Traversal

Step 6: Process node H

## Legend



Complete



Processing



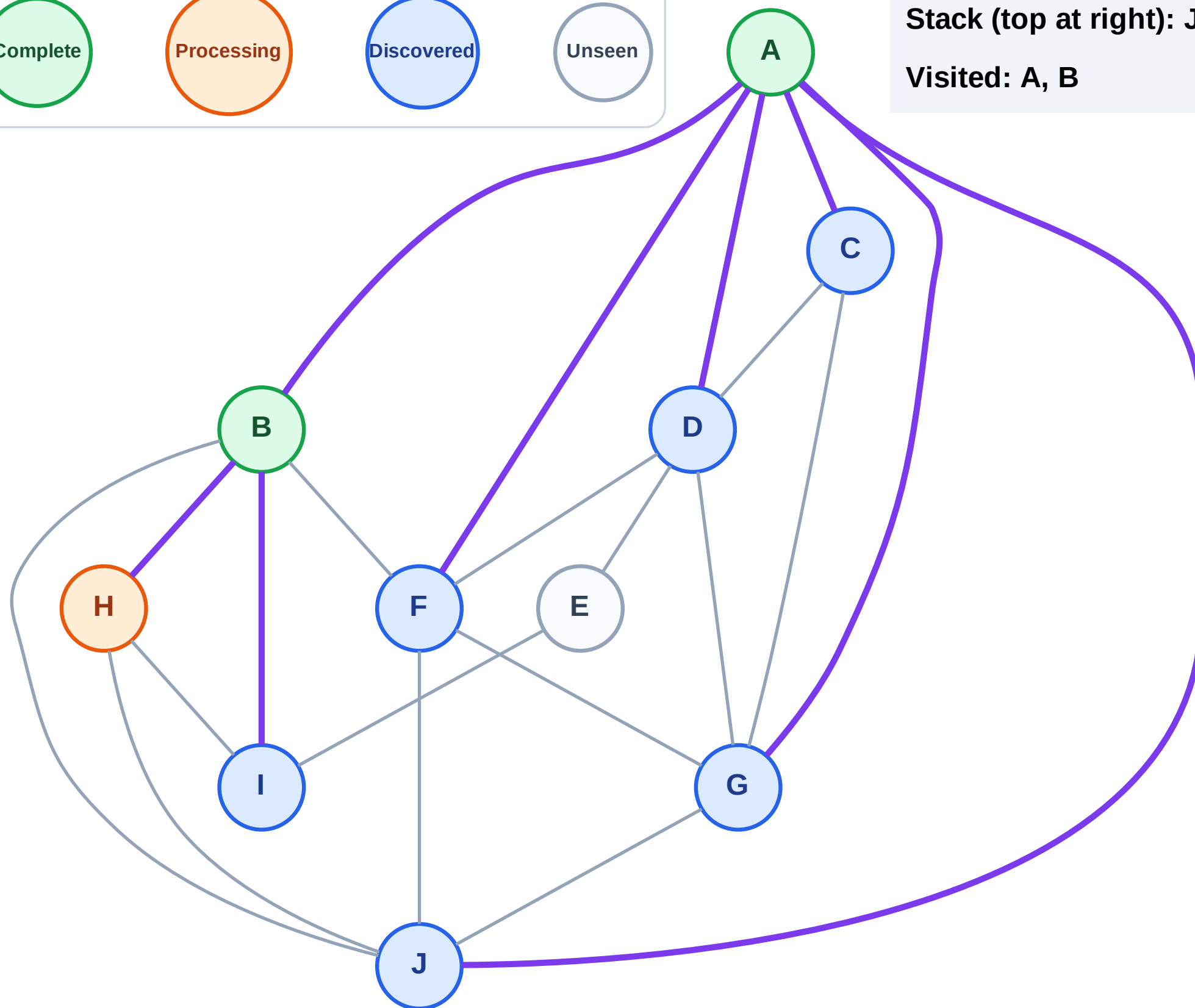
Discovered



Unseen

Stack (top at right): J | G | F | D | C | I

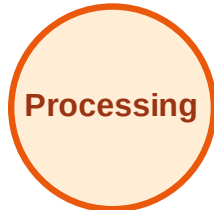
Visited: A, B



# DFS Traversal

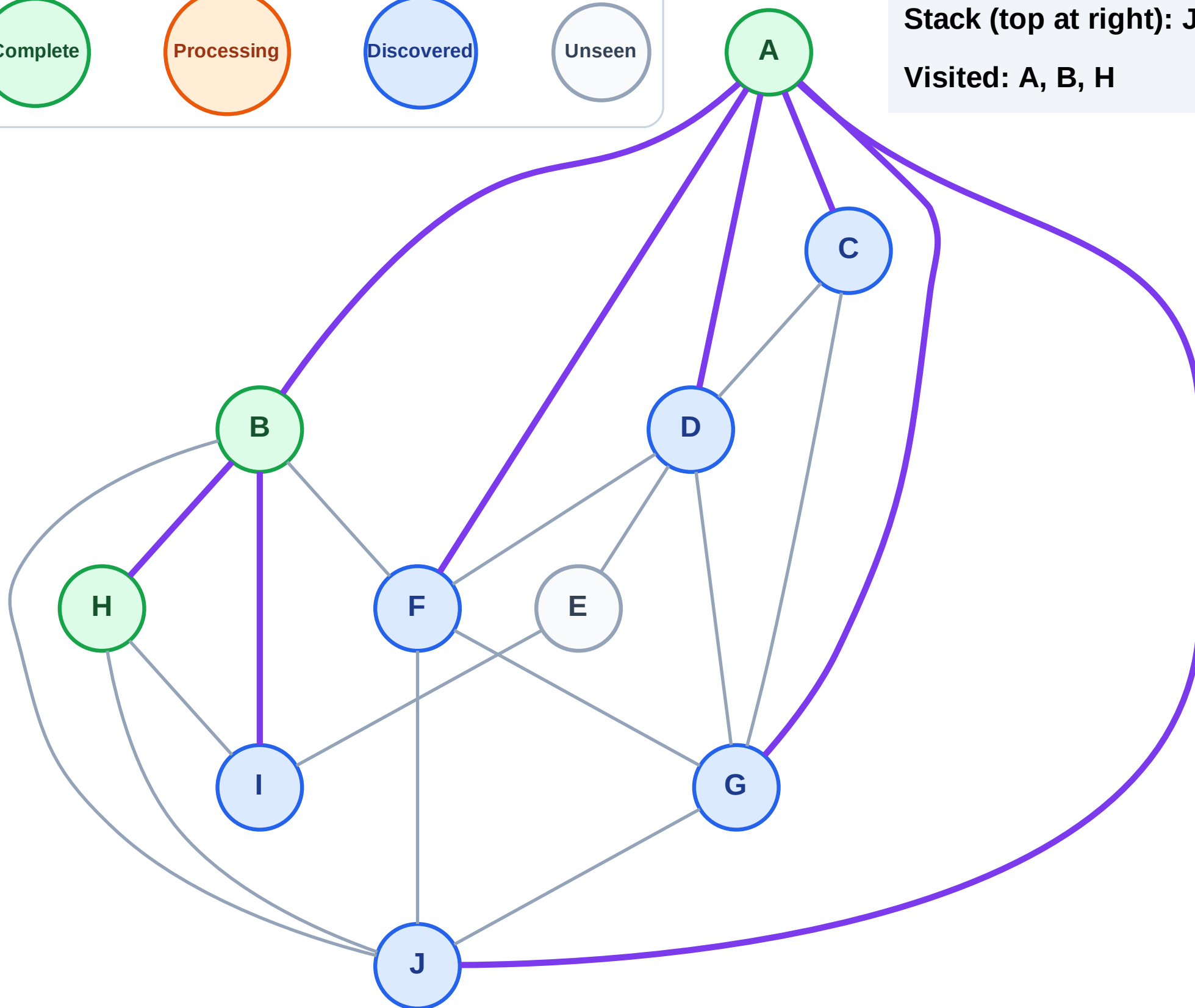
Step 7: No new neighbors from H

## Legend



Stack (top at right): J | G | F | D | C | I

Visited: A, B, H



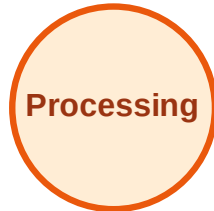
# DFS Traversal

Step 8: Process node I

## Legend



Complete



Processing



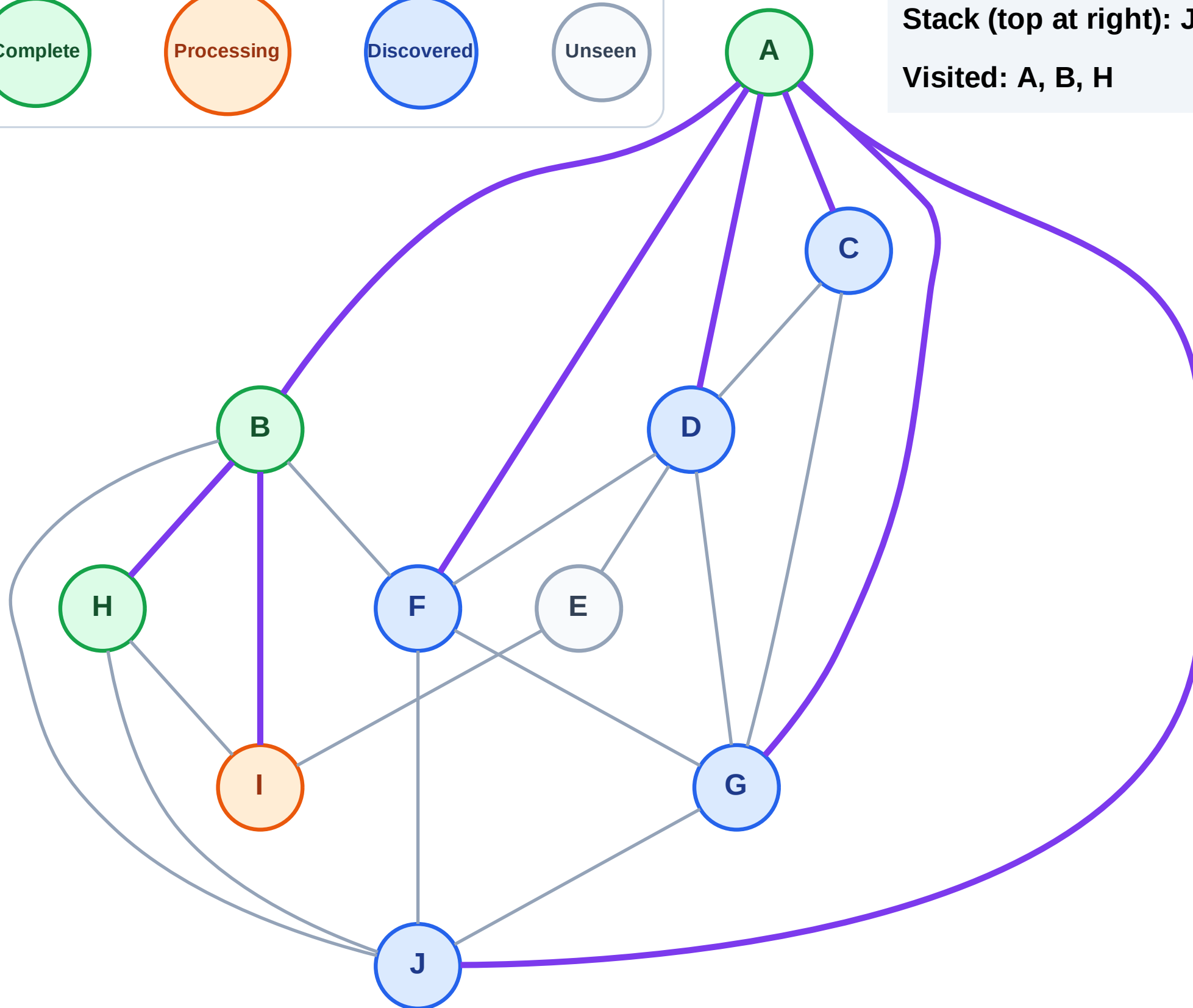
Discovered



Unseen

Stack (top at right): J | G | F | D | C

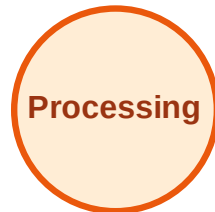
Visited: A, B, H



# DFS Traversal

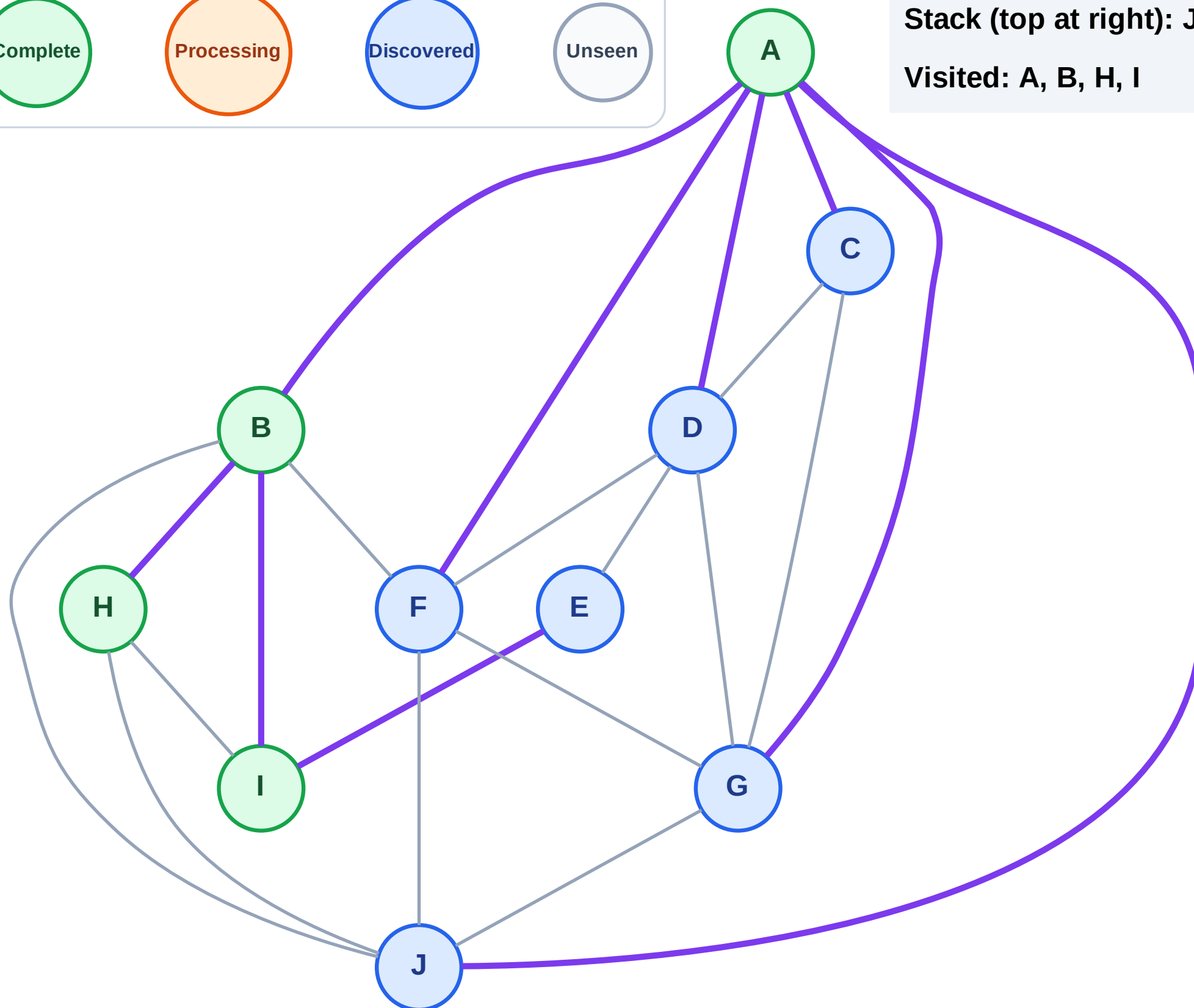
Step 9: Discover E from I

## Legend



Stack (top at right): J | G | F | D | C | E

Visited: A, B, H, I





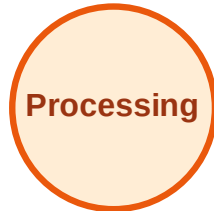
# DFS Traversal

Step 10: Process node E

## Legend



Complete



Processing



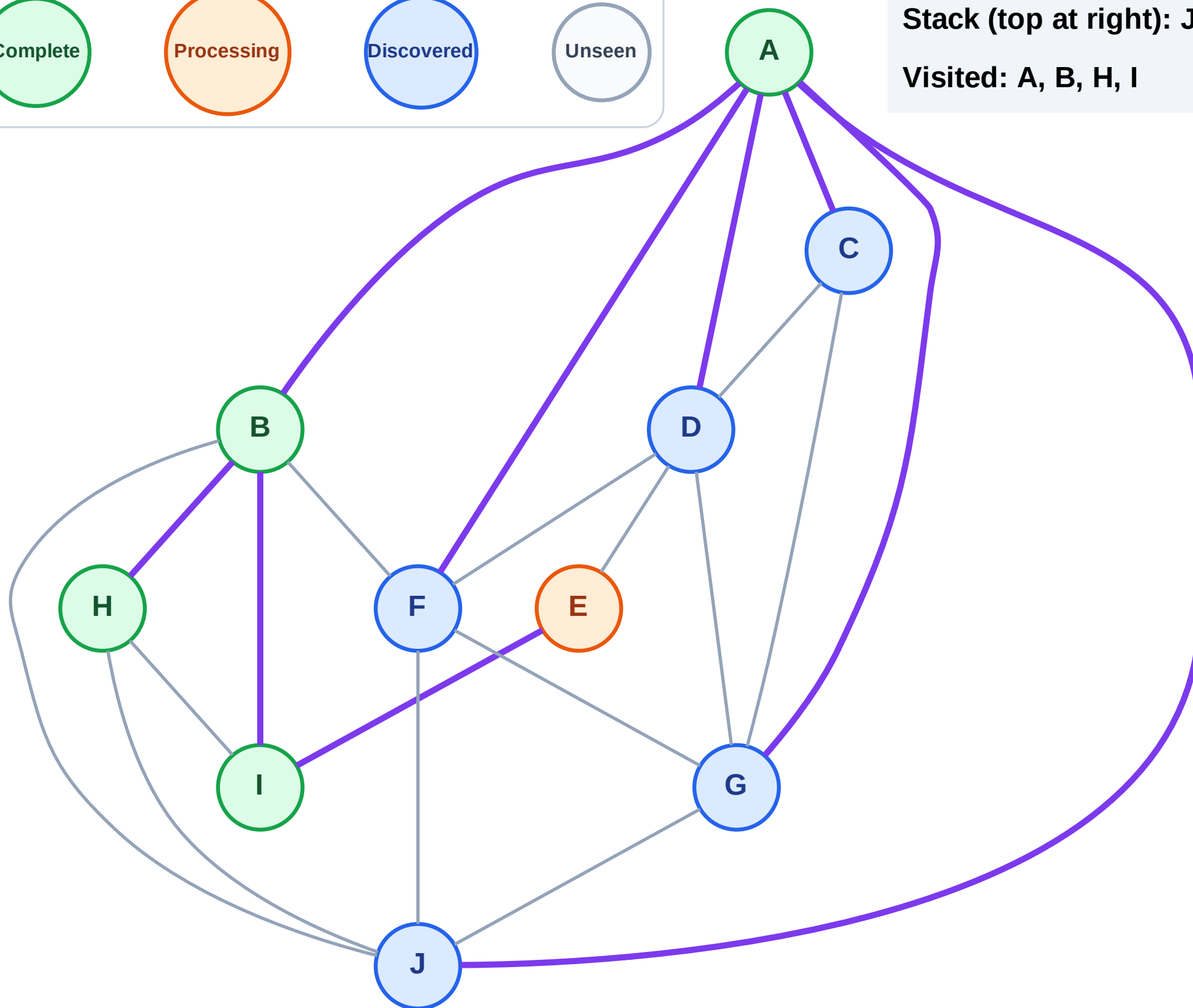
Discovered



Unseen

Stack (top at right): J | G | F | D | C

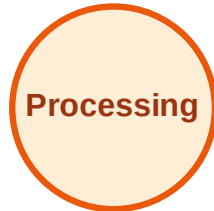
Visited: A, B, H, I



# DFS Traversal

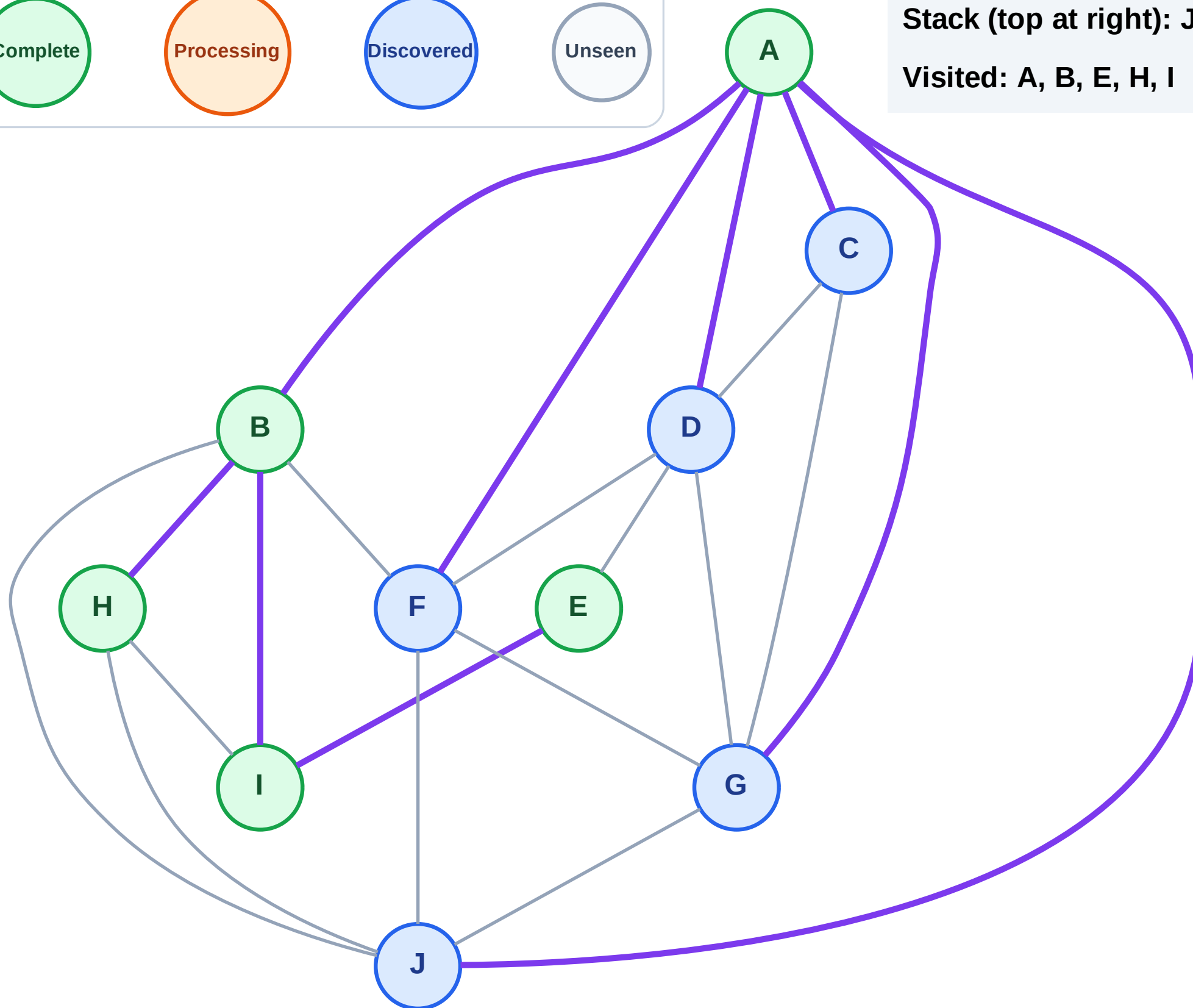
Step 11: No new neighbors from E

## Legend



Stack (top at right): J | G | F | D | C

Visited: A, B, E, H, I

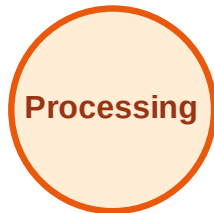




# DFS Traversal

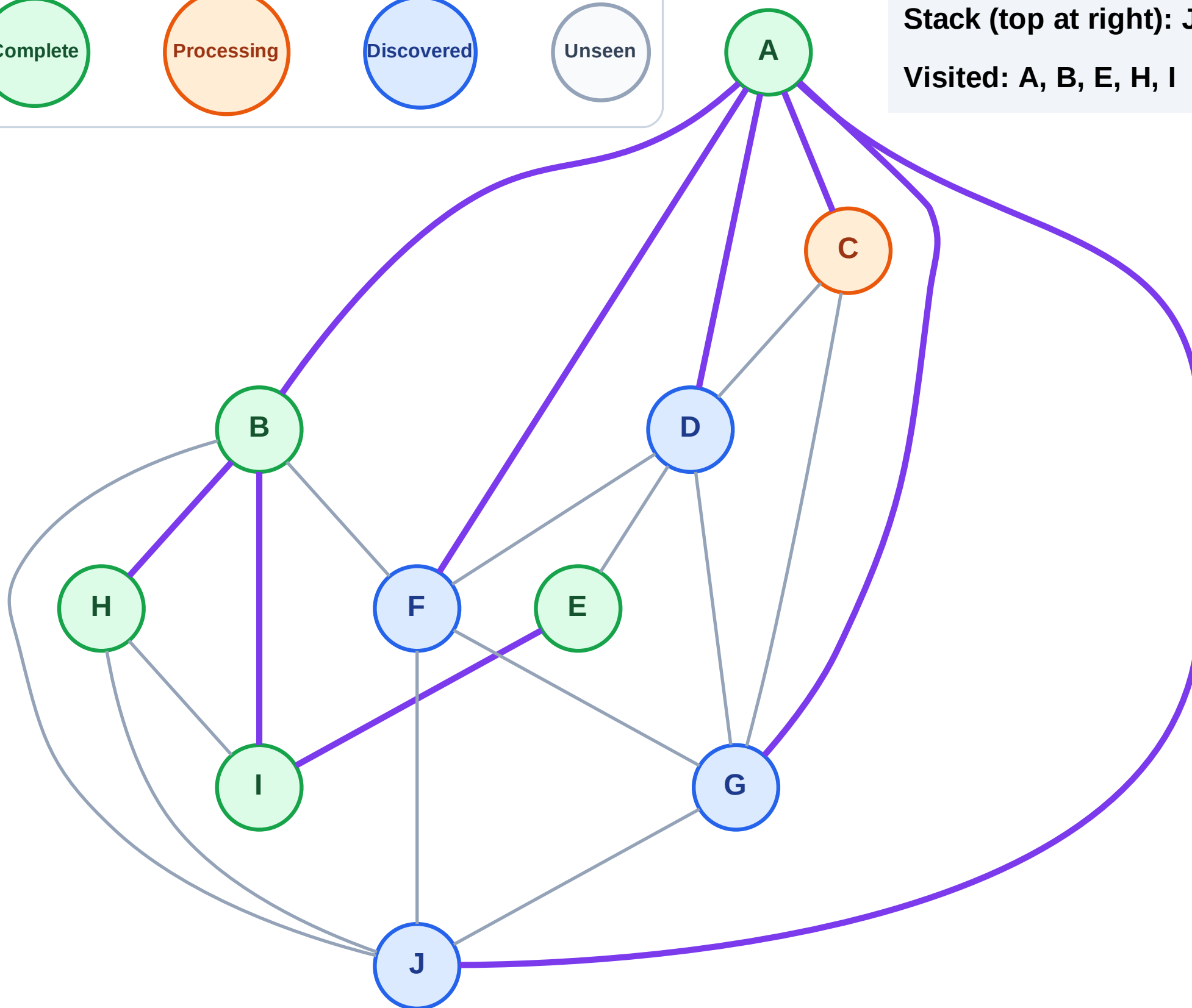
Step 12: Process node C

## Legend



Stack (top at right): J | G | F | D

Visited: A, B, E, H, I



# DFS Traversal

Step 13: No new neighbors from C

Legend

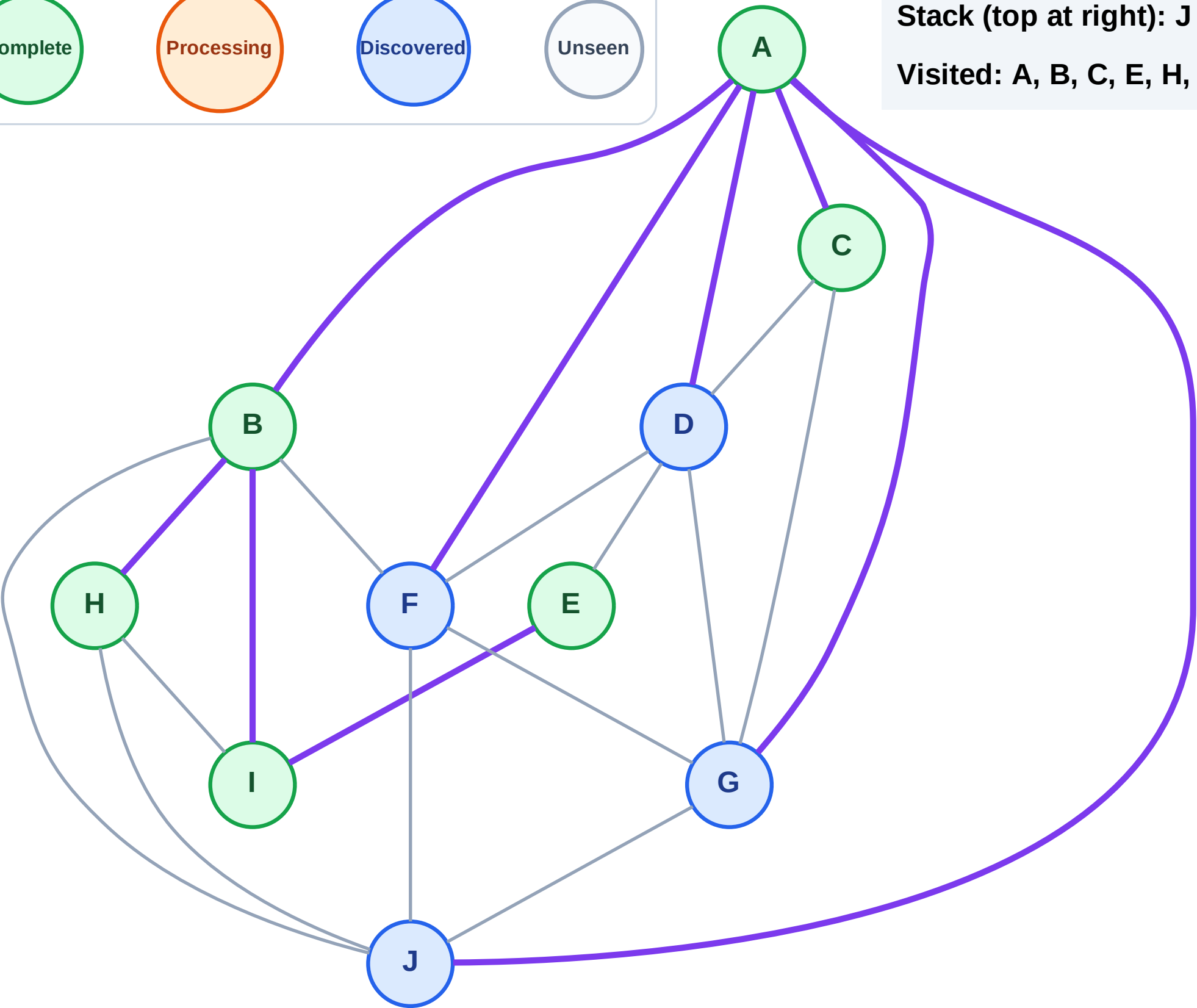
Complete

Processing

Discovered

Unseen

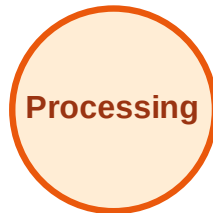
Stack (top at right): J | G | F | D  
Visited: A, B, C, E, H, I



# DFS Traversal

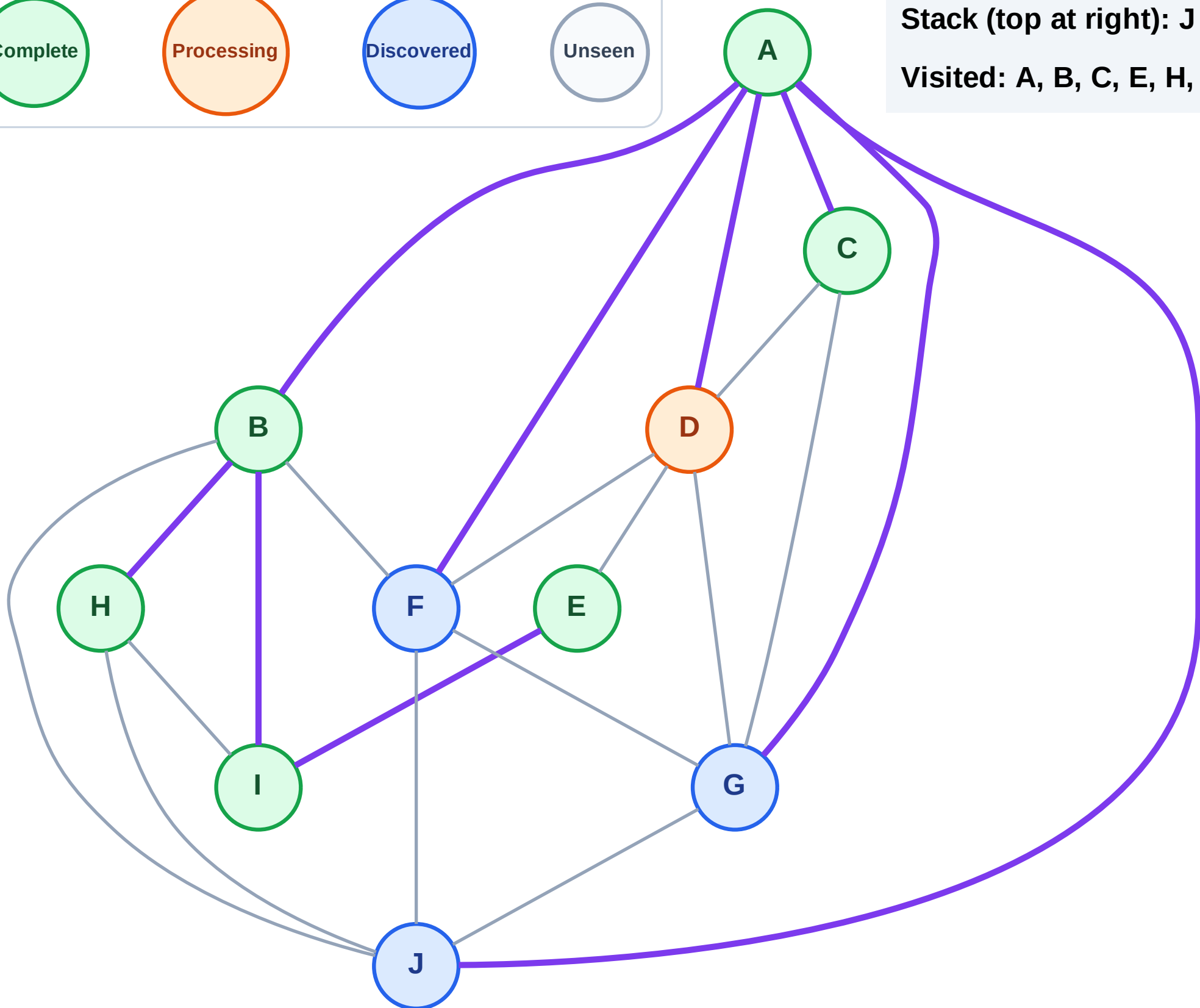
Step 14: Process node D

## Legend



Stack (top at right): J | G | F

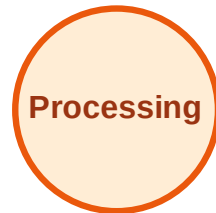
Visited: A, B, C, E, H, I



# DFS Traversal

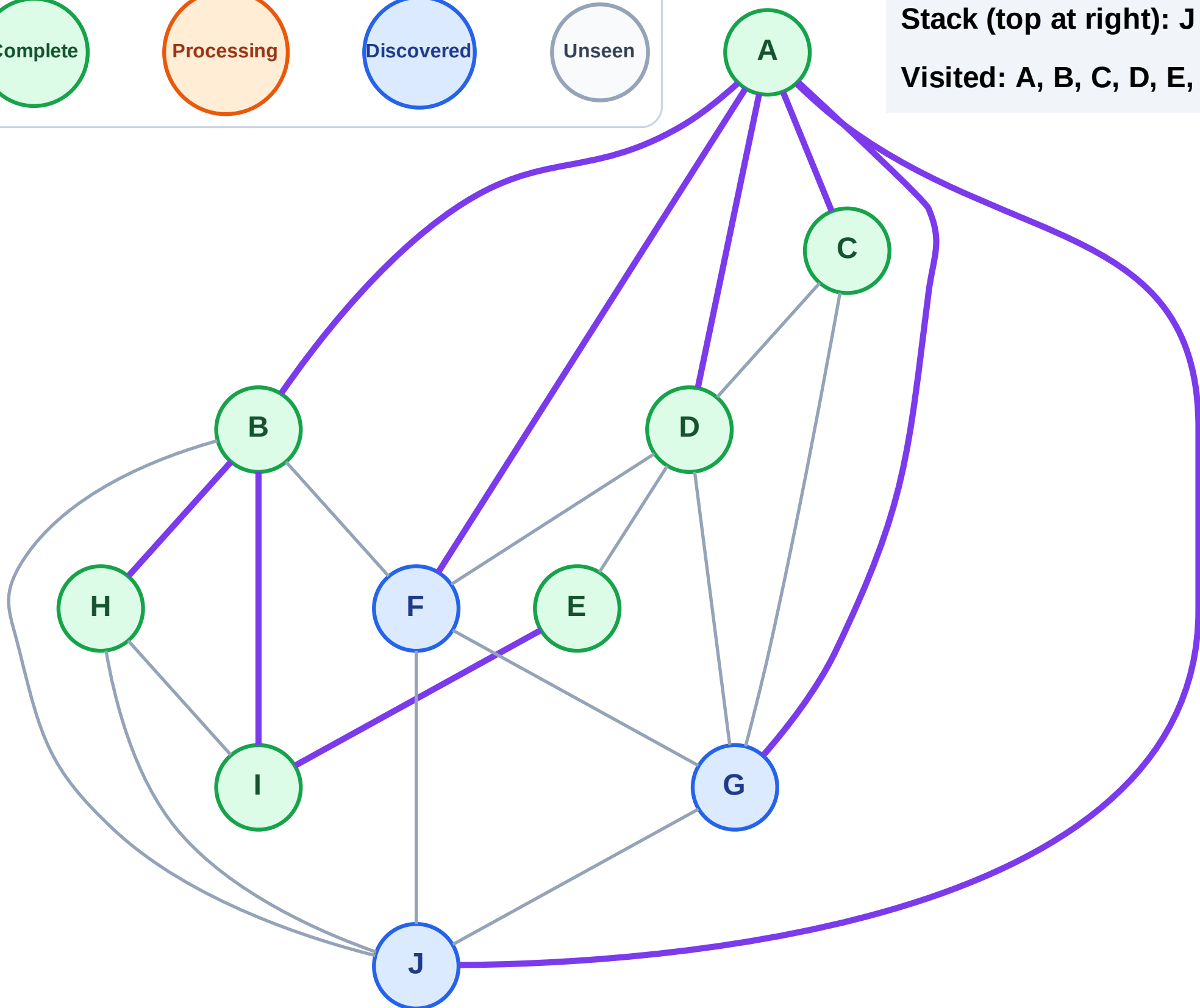
Step 15: No new neighbors from D

## Legend



Stack (top at right): J | G | F

Visited: A, B, C, D, E, H, I



# DFS Traversal

Step 16: Process node F

Legend

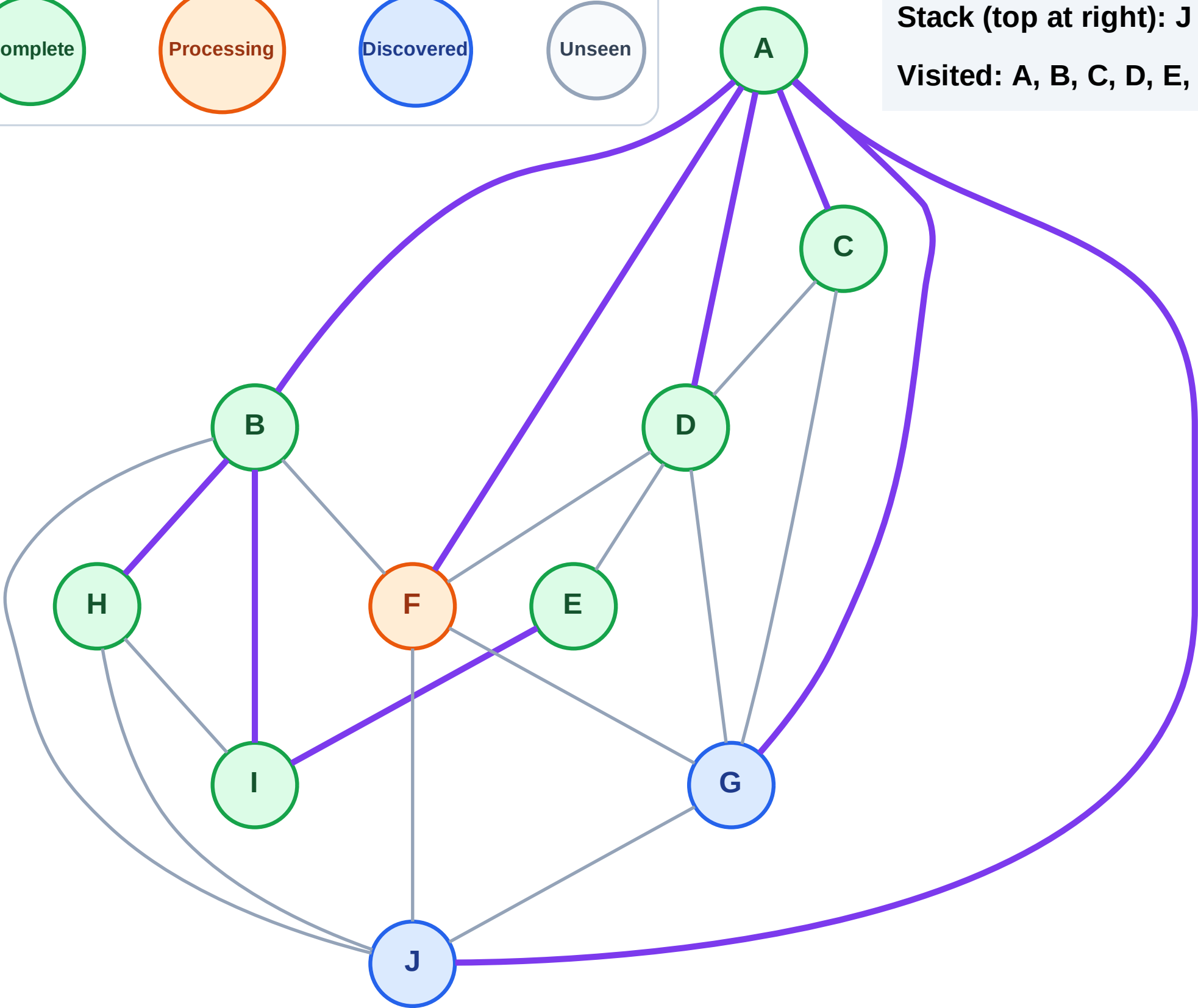
Complete

Processing

Discovered

Unseen

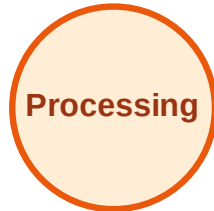
Stack (top at right): J | G  
Visited: A, B, C, D, E, H, I



# DFS Traversal

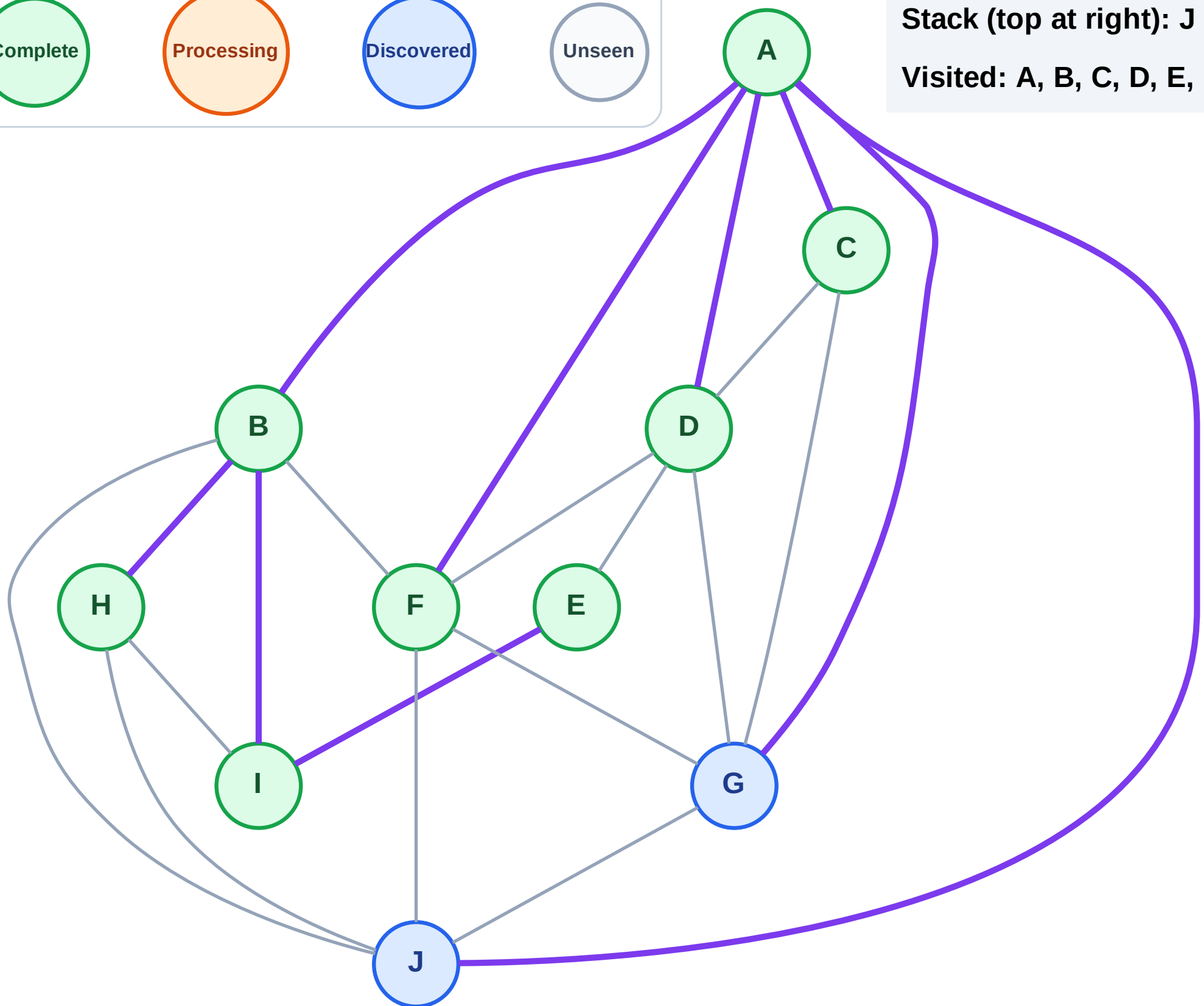
Step 17: No new neighbors from F

## Legend



Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



# DFS Traversal

Step 18: Process node G

Legend

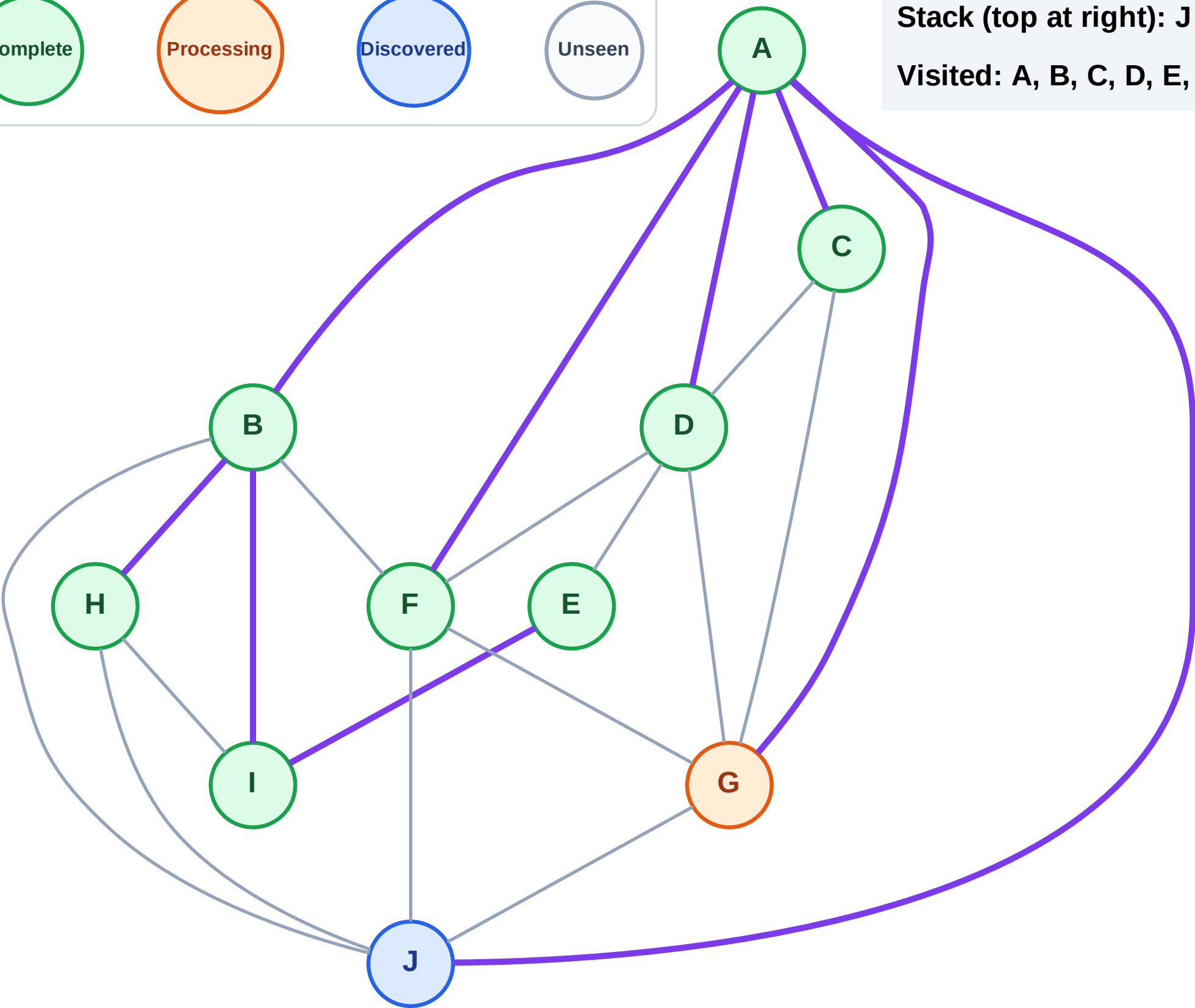
Complete

Processing

Discovered

Unseen

Stack (top at right): J  
Visited: A, B, C, D, E, F, H, I

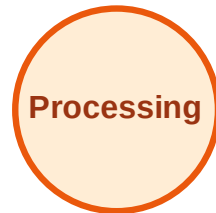




# DFS Traversal

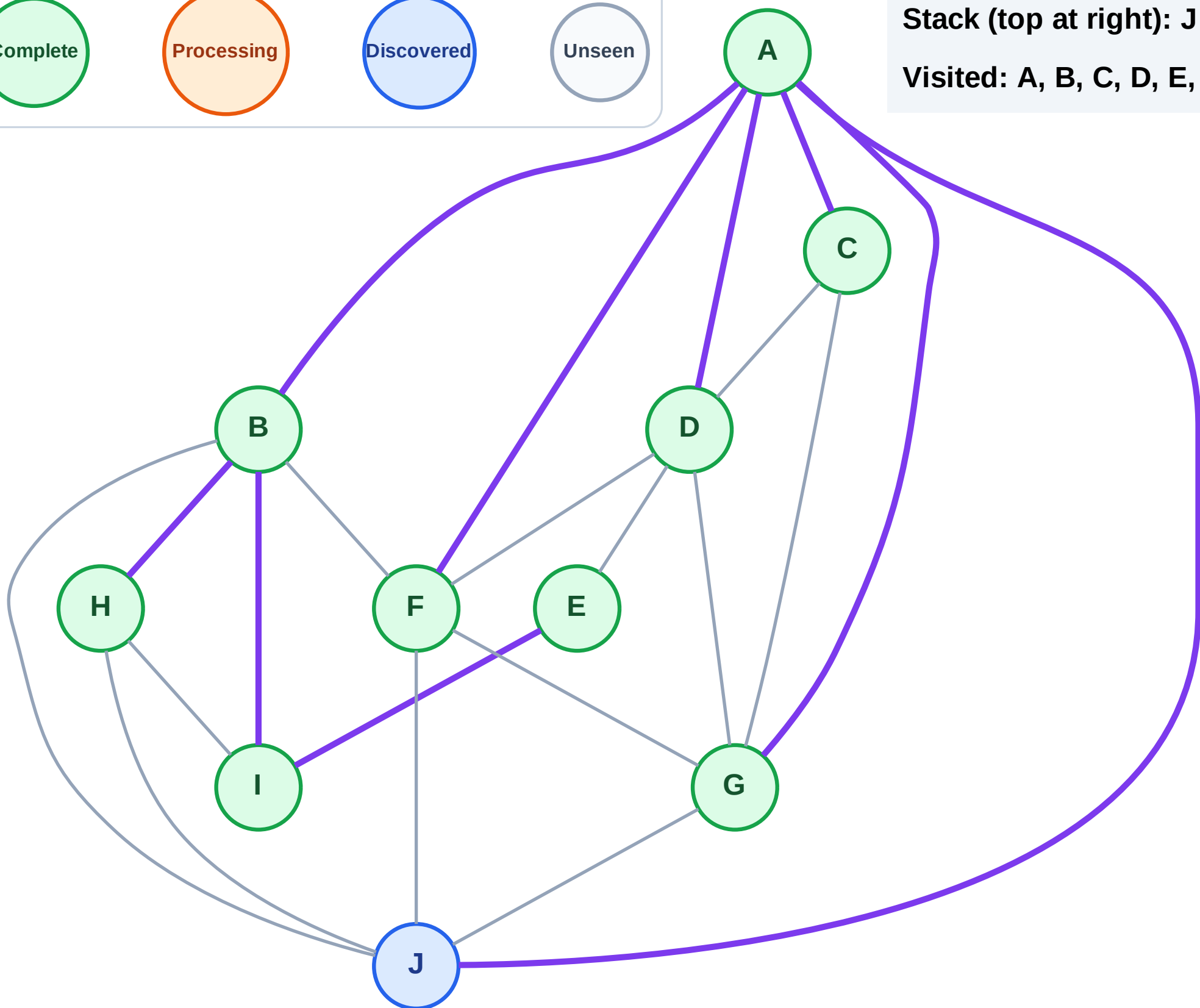
Step 19: No new neighbors from G

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I

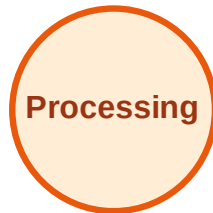




# DFS Traversal

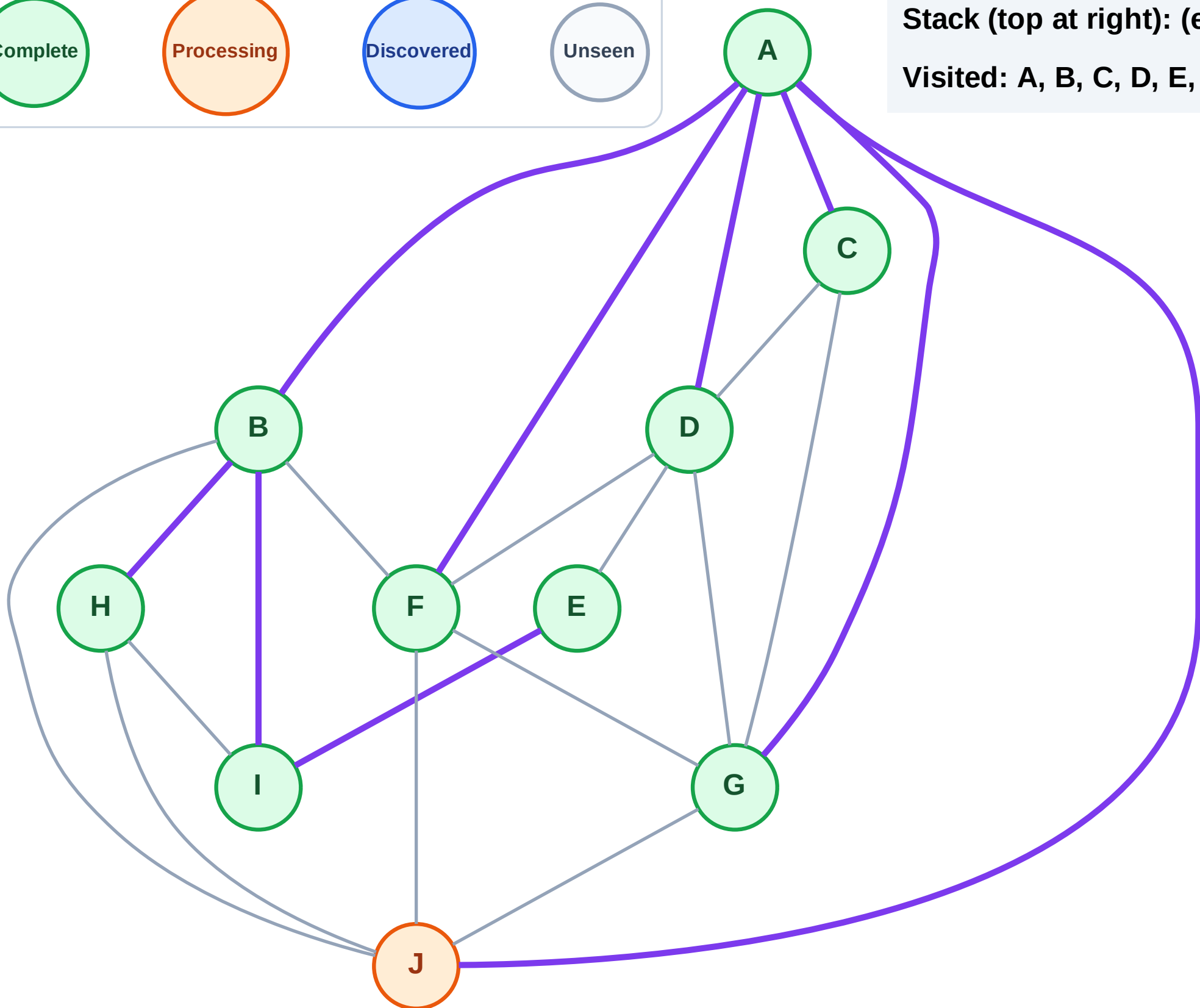
Step 20: Process node J

## Legend



Stack (top at right): (empty)

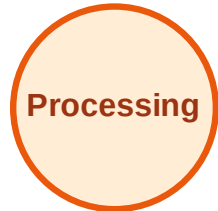
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

